

Design and Simulation of FeFET-Based 8-Bit CIM Array MAC Operations

Dissertation

*Submitted in partial fulfillment of the requirements
for the award of the degree of*

Master of Technology

in

Microelectronics and VLSI

by

**Rahul Kumar Mahato
(24534010)**

Under the Supervision of

Prof. Sudeb Dasgupta

and

Dr. Sourajeet Roy



**Department of Electronics and Communication Engineering
Indian Institute of Technology Roorkee
Roorkee, India
2026**



**Department of Electronics and Communication Engineering
Indian Institute of Technology Roorkee
Roorkee, India**

Candidate's Declaration

I hereby declare that the dissertation entitled “**Design and Simulation of FeFET-Based 8-Bit CIM Array MAC Operations**”, submitted in partial fulfillment of the requirements for the award of the degree of Master of Technology (M.Tech) in Microelectronics and VLSI to the Department of Electronics and Communication Engineering, Indian Institute of Technology Roorkee is an original record of my own work carried out under the supervision of Prof . Sudeb Dasgupta, Professor.

I further declare that this work has not been submitted previously, either in part or in full, for the award of any degree or diploma at this or any other institution.

Rahul Kumar Mahato

Certification

This is to certify that the dissertation entitled “**Design and Simulation of FeFET-Based 8-Bit CIM Array MAC Operations**” submitted by Rahul Kumar Mahato in partial fulfillment of the requirements for the degree of Master of Technology (M.Tech) in Microelectronics and VLSI to the Department of Electronics and Communication Engineering Indian Institute of Technology Roorkee is a bona fide record of work carried out by him under our supervision.

Prof. Sudeb Dasgupta
(Supervisor)

Dr. Sourajeet Roy
(Supervisor)

Acknowledgement

I am grateful to **Prof. Sudeb Dasgupta** and **Dr. Sourajeet Roy**, my thesis supervisors, for their advice and assistance throughout my M.Tech in Microelectronics and VLSI. They provided valuable technical guidance, timely comments, and encouragement during the toughest periods of the project. Their realistic research approach and willingness to assist contributed much to the successful completion of this thesis. I appreciate their approachability and intellectual expertise, which made the research process fruitful and inspiring.

I would also like to acknowledge the faculty members and staff of the Department of Electronics and Communication Engineering, IIT Roorkee, for providing the academic environment and technical facilities necessary for carrying out this research.

I express my heartfelt gratitude to my family for their constant support and understanding. Their patience, encouragement, and belief in me helped me stay focused during difficult phases of this work, and I am deeply thankful for their support throughout this journey.

I am also grateful to all those who contributed to this academic endeavour, directly or indirectly.

Date: 2026-05-28

Rahul Kumar Mahato

Abstract

The rapid growth of data-intensive machine learning workloads has exposed the fundamental limitations of the conventional von Neumann computing architecture, wherein the physical separation between processing and memory units results in excessive data movement, high latency, and dominant energy consumption by memory access rather than computation a phenomenon known as the Memory Wall problem. Compute-in-Memory (CIM) has emerged as a disruptive architectural paradigm that addresses this bottleneck by embedding computation directly within the memory array, thereby eliminating costly data shuttling to external arithmetic units.

This thesis presents the design and simulation of a FeFET-based analog 8×8 Compute-in-Memory array for 8-bit Multiply-Accumulate (MAC) operations, targeting edge AI inference applications. Ferroelectric Field-Effect Transistors (FeFETs), implemented within GlobalFoundries' 28 nm High-K Metal Gate (HKMG) process using silicon-doped hafnium oxide (Si:HfO₂) as the ferroelectric gate dielectric, are employed as non-volatile weight storage elements. The architecture encodes 8-bit inputs via Pulse-Width Modulation (PWM) across four sequential computation cycles and employs a hierarchical 1C-2C-4C capacitive shift-and-add network to combine partial products in the analog domain, reducing the required ADC count by 50% compared to conventional multi-cycle CIM approaches. Each FeFET cell is programmed through a five-cycle bipolar wake-up protocol followed by a state-setting pulse, establishing reliable binary weight storage with a memory window of approximately 1.0–1.5 V. A non-disturbing 100 mV read pulse ensures polarization is preserved throughout inference. The complete architecture is designed and simulated in Cadence Virtuoso using a BSIM Verilog-A compact model that accurately captures ferroelectric P–E hysteresis and threshold voltage bistability.

Simulation results under nominal conditions ($V_{DD} = 1.0$ V, 27°C , typical process corner) demonstrate an energy efficiency of 8.59 TOPS/W, a throughput of 2.13 GOPS, an average power consumption of 98 μW , and a per-cycle latency of 16 ns. These results confirm that FeFET non-volatility, zero standby leakage, and analog in-memory dot-product execution collectively offer a significant advantage over SRAM-based CIM designs.

Table of Contents

Chapter 1: Introduction	1
1.1 The von Neumann Bottleneck and the Memory Wall	1
1.2 The Compute-in-Memory (CIM) Paradigm	1
1.3 Architectural Classification: Digital vs. Analog CIM	2
1.3.1 Digital Compute-in-Memory (DCIM)	2
1.3.2 Analog Compute-in-Memory (ACIM)	3
1.4 The Shift to Emerging Non-Volatile Memories: The Role of FeFETs	3
1.4 Boolean In-Memory Computation Using SRAM	4
1.4.1 Architectural Memory Selection: Volatile vs. Emerging Non-Volatile (FeFET) Technologies	5
Chapter 2: Literature Review	6
2.1 Demonstration of MAC Operation with 28 nm FeFET Crossbar Array	6
2.1.1 Overview and Motivation	6
2.1.2 Key Architecture and Device Design	6
2.1.3 MAC Operation and ADC Results	7
2.1.4 Significance	8
2.2 28 nm HKMG-Based Current Limited FeFET Crossbar Array for Inference Application	8
2.2.1 Overview and Motivation	8
2.2.2 Device Fabrication and Characterization	8
2.2.3 Array-Level MAC and Inference Accuracy	9
2.2.4 Significance	10
2.3 A 28 nm FeFET Compute-in-Memory Macro with 64×64 Array Size and On-Chip 4-Bit Flash ADC	10
2.3.1 Overview and Motivation	10
2.3.2 Macro Architecture	11
2.3.3 Incremental Step Pulse Programming (ISPP) for Variation Reduction	11
2.3.4 DNN Inference Results and Benchmarking	12
2.3.5 Significance	12
2.4 An Efficient FeFET-Based CIM Macro Readout Inspired by the 3T Pixel Array Cell	12
2.4.1 Overview and Motivation	12
2.4.2 FeFET Cell and Variability Mitigation	12
2.4.3 3T-Pixel-Inspired Readout Architecture	13
2.4.4 Performance and Suitability for CNNs and SNNs	14
2.4.5 Significance	14
2.5 Comparative Summary and Research Gaps	15
2.6 Chapter Summary	15

Chapter 3: Design and Implementation of Dual Port 8T SRAM-Based In-Memory Boolean Computation	17
3.1 Motivation for the Proposed Design	17
3.2 Dual Port 8T SRAM-Based Boolean Computation Architecture	17
3.3 Boolean Operation Mechanism.....	18
3.3.1 NAND and NOR Operation.....	19
3.3.2 XOR and XNOR Operation	20
3.4 Transistor-Level Implementation and Simulation Results	21
Chapter 4: FeFET-Based 8-Bit CIM Array MAC Operations.....	24
4.1 Introduction on FeFET Based MAC.....	24
4.2 FeFET Device: Technology, Structure, and Fundamentals	25
4.2.1 What is a FeFET?.....	25
4.2.2 Technology Platform	26
4.2.3 Ferroelectric Polarization and Threshold Voltage Modulation	26
4.3 Key FeFET Array Characteristics.....	27
4.3.1 Non-Volatility	27
4.3.2 Compact Cell Structure.....	27
4.3.3 Programming Voltage Requirements.....	27
4.3.4 Read Disturb Considerations	28
4.3.5 Endurance	28
4.4 FeFET Device Symbol and BSIM Model in Cadence Virtuoso	28
4.5 Weight Programming Protocol (WRITE Operation).....	29
4.5.1 Stage 1: Wake-Up Cycling	29
4.5.2 Stage 2: Writing Logic '1' (LVT State).....	30
4.5.3 Stage 2: Writing Logic '0' (HVT State)	30
4.5.4 Complete Programming Sequence Summary	30
4.6 READ Operation and Sensing Mechanism.....	31
4.7 8×8 FeFET Array Architecture.....	32
4.7.1 Array Organization	32
4.7.2 Weight Mapping Scheme.....	33
4.7.3 2-Bit Weight Grouping and Capacitive Network	33
4.8 8-Bit MAC Operation: Four-Cycle Protocol	34
4.8.1 Input Encoding via Pulse-Width Modulation (PWM).....	34
4.8.2 Four Computation Cycles	34
4.8.3 Parallel Row Activation.....	35
4.9 Detailed Phase-by-Phase Operation.....	35
4.9.1 Phase 1: Pre-Charge.....	35
4.9.2 Phase 2: Discharge (PWM-Driven, FeFET-Dependent)	35
4.9.3 Phase 3: Charge Sharing (1C-2C Capacitive Network).....	36
4.9.4 Phase 4: Output Digitization.....	37
4.10 Four-Cycle Accumulation and Final MAC Output	37

4.10.1 Partial Sums Across Cycles	37
4.10.2 Final MAC Output	37
4.11 ADC Count Comparison: Conventional vs. Proposed Approach	37
4.12 Work Flow	38
4.13 Simulation and Results	39
4.13.1 Simulation in Cadence	39
4.13.2 Timing Waveform.....	41
4.13.3 Voltage Drop Graph.....	45
4.13.4 PVT Analysis	46
4.14 Simulation Results	47
Chapter 5: Conclusion	48
References	50

List of Figures

Figure 1.1: Conceptual View of Von Neumann and Compute-in-Memory Architectures.....	1
Figure 1.2: Type of CIM	2
Figure 1.3: Illustration of Digital and Analog Based CIM Approaches	4
Figure 2.1: Transfer characteristics and CDF distributions comparing 1F and 1F-1T cells, showing the suppression of LVT current variation.....	7
Figure 2.2: MAC operation showing IBL vs. VWL for the 8×8 crossbar segment, and ADC linearity plot.....	8
Figure 2.3: TEM image of the 28 nm HKMG FeFET gate stack and schematic of device states.....	9
Figure 2.4: MLP neural network architecture and inference accuracy vs. time plot showing >97% initial accuracy maintained over 10 ⁸ seconds	10
Figure 2.5: High-level block diagram and schematic of the 64×64 FeFET-CIM macro with peripheral circuits and 4-bit Flash ADC	11
Figure 2.7: Measurement results comparing 1FeFET vs. 1FeFET-1R drain current characteristics, showing current limiting to ~100 nA with reduced spread.....	13
Figure 2.8: Full schematic of the proposed FeFET IMC macro showing the 16×16 segment array, RST/SF readout transistors, inverter comparator, and shared counter	14
Figure 4.1 (Ch.3): Dual Port 8T SRAM-Based In-Memory Boolean Computation Architecture.....	18
Figure 4.2 (Ch.3): Truth table for NAND, NOR, XNOR, and XOR operations	19
Figure 4.3 (Ch.3): Bitline behavior and logic extraction for NAND and NOR operations	20
Figure 4.4 (Ch.3): Bitline behavior and logic extraction for XNOR and XOR operations	21
Figure 4.5 (Ch.3): Transistor-level schematic of the proposed 8T SRAM-based Boolean computation architecture.....	21

Figure 4.6–4.9 (Ch.3): Simulation waveforms for input combinations Q=0/1, SP/SN settings	22–23
Figure 4.1 (Ch.4): Structure of FeFET.....	25
Figure 4.2 (Ch.4): Hysteresis (P-E) curve in ferroelectric materials	26
Figure 4.3 (Ch.4): Proposed 8×8 FeFET Array Architecture	32
Figure 4.4 (Ch.4): Schematic symbol of FeFET	39
Figure 4.5 (Ch.4): Schematic pre-charge circuit.....	39
Figure 4.6 (Ch.4): Schematic of transmission gate 2C-1C and 4C-1C capacitive charge sharing network.....	40
Figure 4.7 (Ch.4): Schematic of a 8×8 FeFET-Based Macro Array.....	40
Figure 4.8 (Ch.4): Waveform of Write 1 operation in FeFET.....	41
Figure 4.9 (Ch.4): Waveform of Write 0 operation in FeFET	41
Figure 4.10–4.15 (Ch.4): Timing waveforms for 8-bit MAC operation.....	42–44
Figure 4.16 (Ch.4): Voltage Drop Graph of Node V4	45
Figure 4.17 (Ch.4): Voltage Drop Graph of Node Vmac	45
Figure 4.18 (Ch.4): PVT Analysis BL	46
Figure 4.19 (Ch.4): PVT Analysis V4	46
Figure 4.20 (Ch.4): PVT Analysis Vmac.....	46

List of Tables

Table 2.1: Comparative summary of reviewed FeFET-CIM works	15
Table 4.1 (Ch.4): FeFET Binary Weight Programming Protocol Summary	30
Table 4.2 (Ch.4): Design Specifications of the Proposed FeFET CIM Array	32–33
Table 4.3 (Ch.4): ADC Count Comparison: Conventional vs. Proposed Approach	38
Table 4.4 (Ch.4): Key Performance Metrics of the Proposed FeFET CIM Array.....	47

Chapter 1

Introduction

1.1 The von Neumann Bottleneck and the Memory Wall

The von Neuman architecture that places a stringent physical and functional separation between the CPU and memory hierarchy has been the foundation on which digital computing has evolved for several decades now. Although modularity facilitates simpler system designs and programming models, it introduces a severe architectural limitation. Specifically, the processor must recurrently read instructions and data from memory, perform calculations, and write back the results.

Due to the scaling of CMOS technology with Moore’s Law, the acceleration of microprocessor clock frequencies and logic densities has taken place at a vastly higher rate than that of the latency and bandwidth improvements of bulk memory. This increasing gap has led to the so-called “Memory Wall.” The energy required to shuttle a single 32-bit operand across the memory bus has become orders of magnitude greater than that required to execute a standard arithmetic operation in 2020.” As workloads like Deep Neural Networks (DNNs) and real-time AI inference are becoming the norm, the continuous data shuttling leads to excessive latency and dominates the system's dynamic power budget. Consequently, the costs associated with the von Neumann bottleneck have shifted, from being merely the cost of latency, now the primary hinderer of energy efficiency.

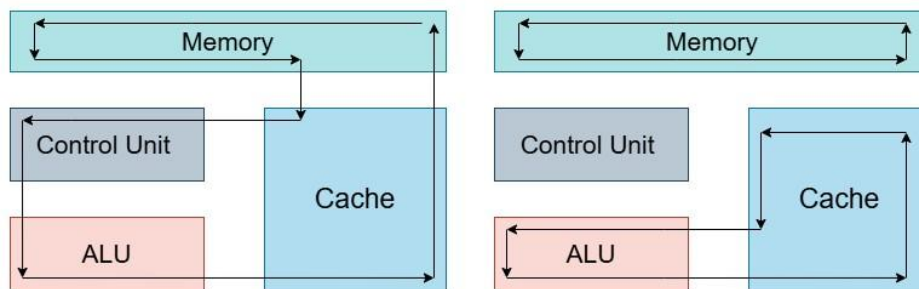


Figure 1.1: Conceptual View of Von Neumann and Compute-in-Memory Architectures.

1.2 The Compute-in-Memory (CIM) Paradigm

To overcome the physical restriction of data movement, Compute-in-Memory (CIM) emerged as a disruptive architectural paradigm. CIM aims to transform the memory macro from a passive storage unit to an almost analog, massively parallel computing unit.

By performing computing inside the memory array itself, CIM essentially allows us to avoid sending a lot of data to external Arithmetic Logic Unit (ALU). This is done by activating multiple memory rows at the same time, and taking advantage of the physical and electrical properties of the memory bitcells and peripheral bitlines. CIM drastically boosts spatial

parallelism, minimizes bottlenecks related to memory access, and substantially decreases the dynamic power dissipation from global interconnect capacitance by processing data in-situ.

1.3 Architectural Classification: Digital vs. Analog CIM

The implementation of Compute-in-Memory largely bifurcates into two types based on working domain namely, Digital CIM (DCIM) and Analog CIM (ACIM).

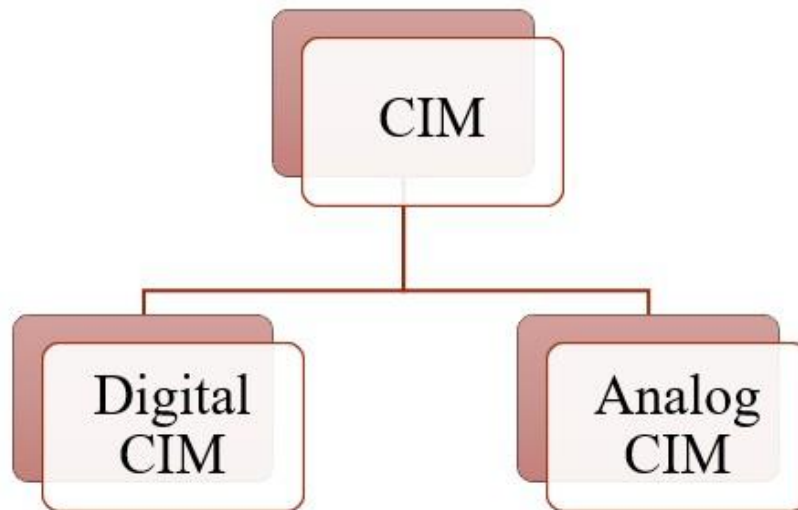


Figure 1.2: Type of Cim

1.3.1 Digital Compute-in-Memory (DCIM)

The operations of DCIM are limited to discrete levels of Boolean logic (0's and 1's), following standard digital CMOS design techniques. In architectures utilizing 6T or 8T SRAM cells, operations are performed by using the discharge behaviour of bitlines to hold exact bitwise operations (NAND, NOR, XOR).

- The DCIM technology is said to have a full-swing deterministic logic, which has excellent static noise margins and is very immune to Process, Voltage and Temperature PVT variations.
- Limitations: The primary drawback of DCIM is area overhead. While excellent for basic Boolean logic, performing complex, multi-bit arithmetic such as Multiply-Accumulate (MAC) operations requires multi-cycle bit-serial processing and massive peripheral digital circuitry (like adders and shift registers), which degrades storage density and throughput.

1.3.2 Analog Compute-in-Memory (ACIM)

The ACIM utilizes the persistent analog behavior of memory devices to conduct arithmetic computing in parallel. In current-domain ACIM, the analog voltage (V) is used to generate the variable conductance (G) of the memory cell's stored value. According to Ohm's law the current is generated by the cell ($I=VG$) According to Kirchhoff's Current Law (KCL), these individual currents combine together in the vertical shared bitlines, which computes a dot-product operation in parallel and in one step.

- Advantages: ACIM achieves large vector-matrix multiplications in a highly parallel manner in $O(1)$ time complexity to obtain enormous energy-delay products essential for AI accelerators.

1.4 The Shift to Emerging Non-Volatile Memories: The Role of FeFETs

Static random-access memory (SRAM) has an excellent, logic-based, and CMOS-compatible foundation for implementing DCIM. SRAM's volatility and high static leakage power, with its overly large physical cell size, limits its scalability for huge, analog matrix operations. For contemporary applications of ANM, the field is now moving towards new Non-Volatile Memories (NVMs) capable of dense multi-bit storage.

Of these technologies the Ferroelectric Field-Effect Transistor (FeFET) stands out as very promising for in-memory computation. A FeFET has a ferroelectric material (HfO_2) in its gate stack (in the case of a standard MOSFET). When specific voltage pulses are applied to the gate, the polarization state of the ferroelectric layer is modified, resulting in a non-volatile shift in the transistor threshold voltage (V_{th}).

The ability to change part of the polarization is important for FeFETs Such a capability allows for a single FeFET device to act as a multi-level cell (MLC) which is capable of storing various analog conductance states that can accurately emulate the multi-bit synaptic weights that are necessary for neural networks. By using FeFETs in a crossbar array, the design can perform 8-bit analog MAC operations with unprecedented density and essentially zero static leakage power. This thesis investigates the architectural evolution from using regular SRAM for simple Boolean logic to employing the unique device physics of FeFETs for high-accuracy arithmetic computation.

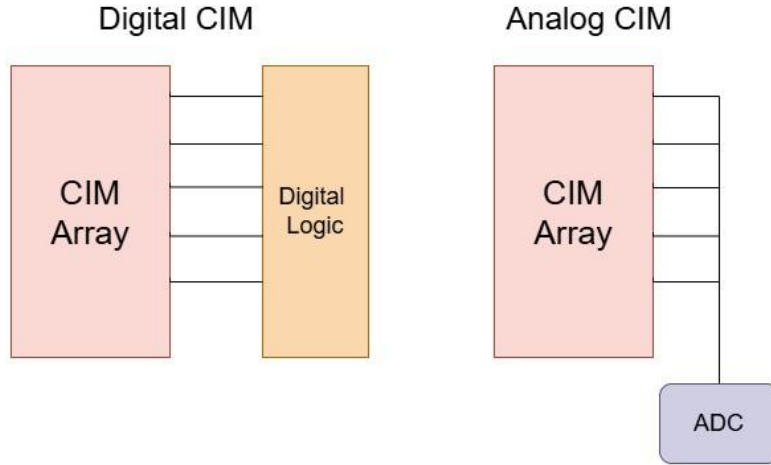


Figure 1.3: Illustration Of Digital and Analog Based Cim Approaches.

1.4 Boolean In-Memory Computation Using SRAM

Boolean Compute-in-Memory (CIM) architectures implement basic logic operations directly in the memory macro itself. This approach eliminates the latency and energy cost of continuously sending data to a remote Arithmetic Logic Unit (ALU). This paradigm allows for in-situ logic evaluation using the electrical behaviors of the memory bitcells and the memory peripheral circuitry rather than treating the memory array merely as a data store.

SRAM being faster than a nanosecond and stable is an ideal medium for implementing bitwise Boolean computation, L1/L2 caches and register files (the blocks of combinational logic) being found in 10s of millions on modern chips. Moreover, SRAM can be fabricated using standard CMOS technology allowing the integration of CIM functionalities easily without the use of exotic materials or process nodes.

SRAM-based logic execution of computation is usually triggered by MWL activation. When two or more wordlines are asserted at the same time, memory cells compete to discharge a precharged, shared bitline. The total voltage on the bitline is computed for each row in the DRAM array. This overall signal is then processed by peripheral sense amplifiers without the need of reading the individual operands serially and routing them.

Nevertheless, there are significant circuit-level implications of retrofitting standard SRAM arrays for CIM. An advantage of the 6T SRAM organization is the coupling of write and read paths. The simultaneous activation of multiple wordlines can subject the internal storage nodes to significant vulnerabilities, including half-select and read-disturb. The reliability of the logic evaluation is impaired due to decreased Static Noise Margin (SNM) and capacitive bitline coupling. Therefore, strong 6T SRAM-based computation is realized by accurate sizing of access transistors and tight sensing strategies. SRAM is still one of the best technologies to implement localized, fast logic despite the challenges.

1.4.1 Architectural Memory Selection: Volatile vs. Emerging Non-Volatile (FeFET) Technologies

The underlying memory technology chosen depends on the characteristics of the computational workload. Memory technologies can be categorized into two major groups – volatile (SRAM, DRAM) and non-volatile (RRAM, PCM, FeFET) – each of which has its own architectural trade-offs.

- **The Case for Volatile Memories (High-Speed Logic)**

Volatile memories, especially SRAM, are unmatched for local, bitwise Boolean operations with extremely low latency and continuous data updates. Volatile cells have symmetric, high-speed read and write profiles, essential for iterative logic executions.

More importantly, SRAM has infinite write endurance, which signifies that during different types of cycling, there is no physical wear-out mechanism in the cells. By contrast, the asymmetric write latencies and large programming energies of many conventional NVMs are coupled with limited lifetime endurance, which renders them unsuitable for dynamic Boolean variables. Moreover, SRAM demonstrates exceptionally deterministic electrical properties, unlike the resistance drift and retention degradation seen in resistive NVMs.

- **The Case for FeFETs (High-Density Analog Arithmetic)**

Although SRAM is the one and only circuit architecture capable of executing fast Boolean logic, the large physical footprint and binary nature of SRAM make it very inefficient for storing large weight matrices and storing high precision (8-bit MAC) multiply-accumulate (MAC) for neural networks.

To address this gap, **Ferroelectric Field-Effect Transistors (FeFETs)** have become a disruptive NVM technology for analog CIM applications. A FeFET incorporates a ferroelectric hafnium oxide (HfO₂) layer into the CMOS gate stack. It offers many benefits which are very appealing than SRAM.

- **Analog Multi-Level Cell (MLC) Capability:** By modulating the polarization of the ferroelectric layer with precise gate pulses, a single FeFET can exhibit a continuous range of threshold voltages (V_{th}). This allows one compact device to store multi-bit synaptic weights (e.g., 8-bit precision), vastly increasing array density compared to SRAM.
- **Voltage-Driven Operation:** FeFETs are more energy-efficient than PCM or STT-MRAM in terms of write operation as they don't require energy-intensive current pulses but are programmed with voltage-driven electric fields.
- **Kirchhoff-Based MAC Execution:** FeFETs have programmable conductances (G) in a crossbar array. The application of an input activation voltage (V) on the wordline results in a proportional drain current: ($I=V G$) for a FeFET. These currents also combine in accordance with Kirchhoff's Current Law along bitlines, enabling massively parallel MAC operations to occur in a single analog step (i.e., $O(1)$ time complexity).

Chapter 2

Literature Review

The explosive growth of edge artificial intelligence (AI) applications is creating the demand for energy-efficient hardware accelerators that can do deep neural network (DNN) inference free of the latency and power overhead of traditional von Neumann architectures. Compute-in-memory (CIM) is a new paradigm in which computation takes place in memory array and hence movement of data is reduced significantly. Among the various emerging non-volatile memory (eNVM) technologies explored for CIM, ferroelectric field-effect transistors (FeFETs) based on hafnium oxide (HfO₂) have attracted significant research attention owing to their CMOS compatibility, high on/off current ratio, low read/write energy, and scalability into advanced technology nodes. This chapter reviews four key studies that collectively demonstrate the progression of FeFET-based CIM technology from fundamental multiply-accumulate (MAC) operation on 28 nm crossbar arrays, through current-variation mitigation strategies, to fully integrated macro designs with on-chip analog-to-digital converters (ADCs) and novel readout architectures.

2.1 Demonstration of MAC Operation with 28 nm FeFET Crossbar Array

2.1.1 Overview and Motivation

De et al. (2022) presented one of the first experimental demonstrations of a linear multiply-accumulate (MAC) operation using a crossbar memory array based on 28 nm high-k metal gate (HKMG) CMOS technology integrating FeFET devices fabricated at GlobalFoundries [De et al., IEEE Electron Device Letters, Vol. 43, No. 12, 2022]. A central challenge addressed in this work was the device-to-device variation in drain current, particularly in the low-threshold voltage (LVT) state, which had previously been identified as a fundamental bottleneck to reliable array-level operation. In an effort to mitigate this issue, the authors proposed a transistor known as MCL, which had its source line (SL) cascoded with that of each 8×8 FeFET crossbar partition. This method maintains the simplicity of the macro design while greatly suppressing the spread of LVT current.

2.1.2 Key Architecture and Device Design

The FeFET crossbar array employed an 8 nm silicon-doped HfO₂ ferroelectric layer with a ~1 nm SiO₂ interfacial layer integrated into the gate stack of a standard MOSFET. Devices were operated in binary states: a low-VT (programmed, logic '1') state achieved by a +4.5 V, 500 ns gate pulse, and a high-VT (erased, logic '0') state set by a −5 V pulse. The memory array was divided into 8×8 segments to limit bitline current accumulation errors and reduce word-line voltage swing effects. Critically, the access transistors, current-limiting transistors, and a thermometer-code current-mode ADC were all co-fabricated on

the same 300 mm wafer, representing a significant step toward a fully integrated FeFET-CIM macro.

The 1F-1T cell (FeFET with current-limiting transistor) achieved a mean LVT bitline current of approximately 100 nA with a standard deviation below 3 nA, compared to ~ 0.8 μ A variation in the standalone 1F cell with a mean of ~ 7 μ A. This remarkable decline in variability enables dependable MAC summation over a multitude of activated rows.

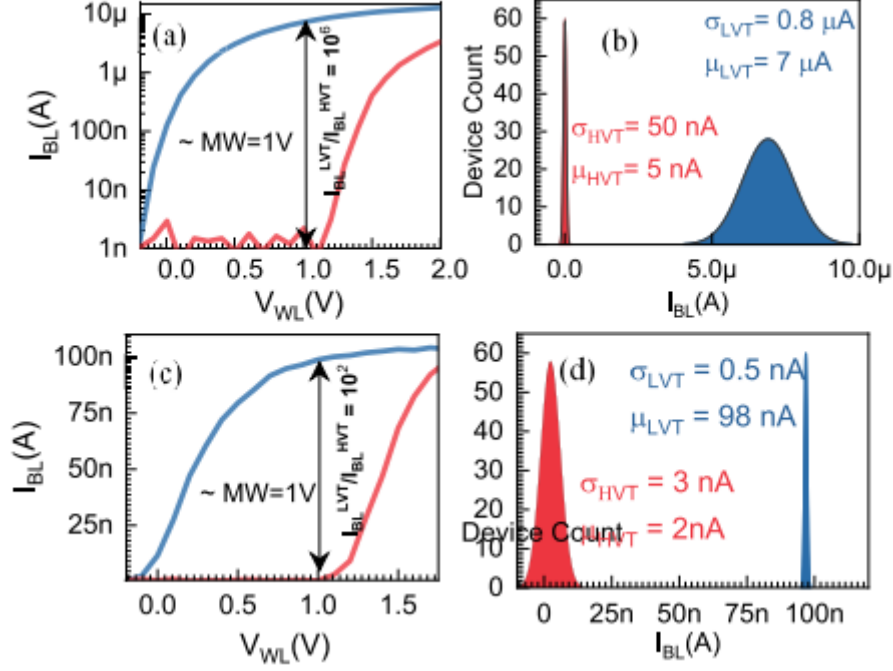


Figure 2.1: Transfer characteristics and CDF distributions comparing 1F (standalone FeFET) and 1F-1T (FeFET + current-limiting transistor) cells, showing the suppression of LVT current variation Fig. 2 from De et al. (IEEE EDL, 2022)

2.1.3 MAC Operation and ADC Results

The MAC operation was confirmed for more than 20 segments across 300 mm wafer crossbar arrays, showing a 100% yield with stable and linear accumulation of bitline current with each additional row activated. The code corresponding to 100 nA reference current thermometer-code ADC was able to reliably convert the accumulated bitline current into a digital output with Pearson's correlation coefficient to close to 1. Data retention at room temperature shows no degradation up to 5×10^4 sec for all activation levels. Extrapolation indicates retention of ten years which is important for inference-engine applications with rare or offline weight updates.

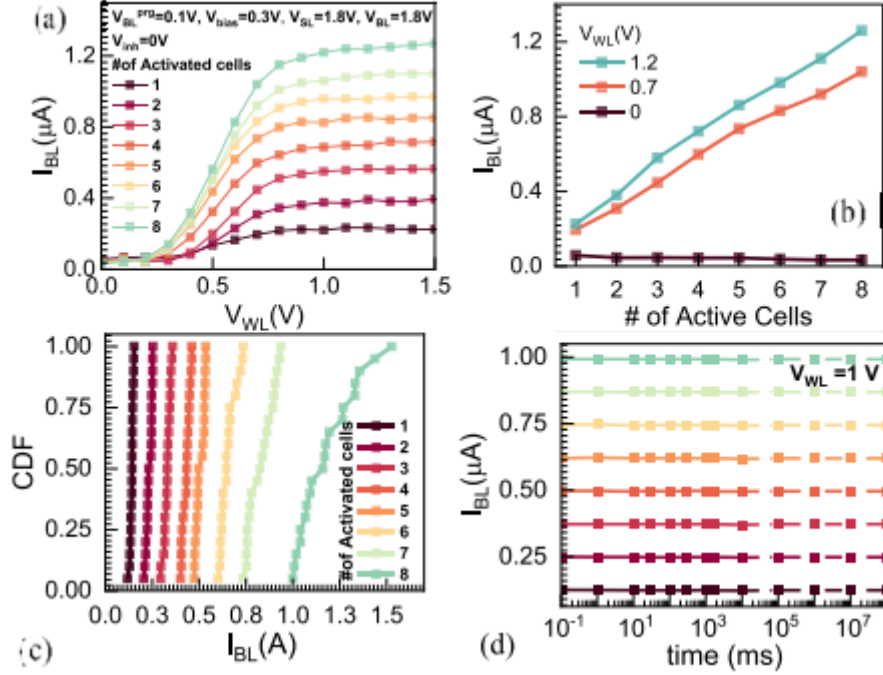


Figure 2.2: MAC operation showing IBL vs. VWL for the 8×8 crossbar segment, and ADC linearity plot Fig. 4 from De et al. (IEEE EDL, 2022)

2.1.4 Significance

This work established the feasibility of fabricating a complete FeFET-CIM unit array, peripheral circuits, and ADC on a single wafer without any post-process modifications, and demonstrated that device-level variation can be controlled at the circuit level through current limiting. The 28 nm HKMG platform provided by GlobalFoundries forms the common fabrication baseline for the subsequent works reviewed in this chapter.

2.2 28 nm HKMG-Based Current Limited FeFET Crossbar Array for Inference Application

2.2.1 Overview and Motivation

Building directly on the prior MAC demonstration, De et al. extended their investigation to a system-level evaluation of current-limited FeFET crossbar arrays for neuromorphic inference tasks [De et al., IEEE Transactions on Electron Devices, Vol. 69, No. 12, 2022]. The core innovation in this work was the use of a 1F-1R structure an external resistor shunted with the drain terminal (or equivalently, shunting the column of the FeFET array) as a current limiter rather than a transistor-based limiter. A resistance of 2 MΩ was found to sufficiently suppress LVT-state current variation originating from random distribution of ferroelectric domains, while the large gate voltage operation window further contributed to variability reduction across the word lines.

2.2.2 Device Fabrication and Characterization

Fabricated in GlobalFoundries' 28 nm HKMG process on 300 mm wafers, the FeFETs incorporated an 8 nm Si-doped HfO₂ ferroelectric layer. Prior to READ-WRITE

characterization, devices were conditioned by 50 wake-up pulses (alternating +4.5 V / -5 V, 500 ns each) to stabilize the ferroelectric orthorhombic phase responsible for switchable polarization. The 1F-1R structure reduced the LVT-state current spread significantly: although the mean LVT current decreased, the standard deviation dropped sharply, ensuring more uniform bitline current accumulation and reducing ADC conversion errors. The memory window remains stable upto 10^3 write cycles during the endurance test after which it starts deteriorating gradually. The limit of full closure is attained around 5×10^5 cycles. Up to 10^4 seconds, retention at 85°C was stable.

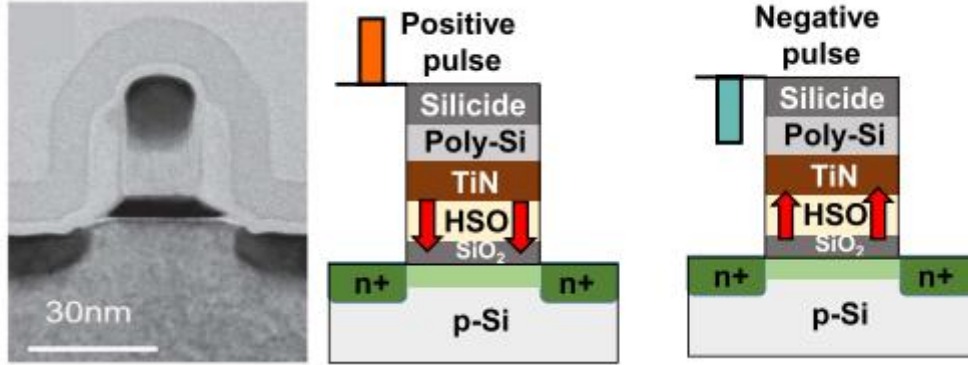


Figure 2.3: TEM image of the 28 nm HKMG FeFET gate stack (showing HfO₂ ferroelectric layer) and schematic of device states Fig. 1 from De et al. (IEEE TED, 2022)

2.2.3 Array-Level MAC and Inference Accuracy

The 8×8 crossbar array showed very small leakage current (< 1 nA with all cells erased). Also, the MAC operations done over 20 segments on a 300 mm wafer did not overlap nor shift original bitline current distributions (stable and reproducible). System-level validation through neuromorphic simulation concentration of handwritten digits on MNIST dataset using a multi-layer perceptron (MLP) neural network (architecture: 400-input, 100-hidden, 10-output). The network underwent offline training to satisfy the constraints of FeFET synapses endurance. The current-limited FeFET crossbar array obtained an inference accuracy of more than 97% and finished with an accuracy of more than 95% after 10^8 seconds without retraining thus effectively demonstrating decade-long retention adequacy for edge inference.

In comparison, a crossbar array composed only of independent FeFETs (without current limiting) did not converge and achieved only 9.8 percent accuracy which was essentially at chance for 10-class classification. The above point reflects the need for variation suppression for analog CIM applications.

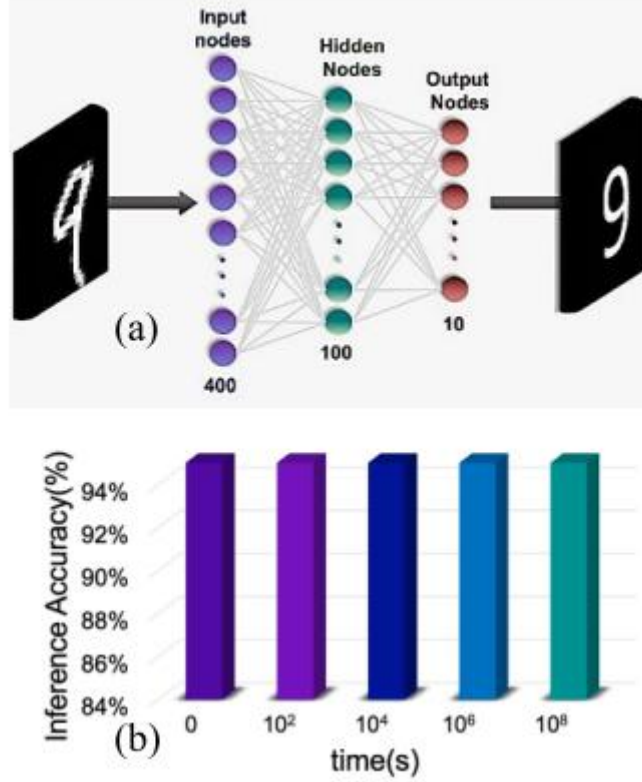


Figure 2.4: MLP neural network architecture and inference accuracy vs. time plot showing >97% initial accuracy maintained over 10^8 seconds Fig. 6 from De et al. (IEEE TED, 2022)

2.2.4 Significance

As far as we are aware, this study is the furthest advanced system-level demonstration of the 1F-1R FeFET crossbar concept, confirming that a simple passive resistor can sufficiently compensate for device variability to enable software-comparable inference accuracy on real classification benchmarks. The 28 nm technology node and GlobalFoundries platform comparisons (benchmarked against FeRAM and FeFET works at 130–180 nm nodes) highlight the density and voltage advantages of the scaled HKMG process.

2.3 A 28 nm FeFET Compute-in-Memory Macro with 64×64 Array Size and On-Chip 4-Bit Flash ADC

2.3.1 Overview and Motivation

Garg et al. (2026) presented the first FeFET-CIM prototype macro at the full chip level, fabricated in the GlobalFoundries 28 nm HKMG process and reported in IEEE Solid-State Circuits Letters [Garg, Jia, Phadke, and Yu, IEEE SSCL, Vol. 9, 2026]. While prior works demonstrated MAC operations on small 8×8 arrays and bare dies, this work scaled to a 64×64 FeFET crossbar array (4 kb capacity) with eight integrated 4-bit Flash ADCs, configurable input and weight precision, and full peripheral circuitry addressing the practical requirements of DNN inference at the edge. A key motivation was to surpass the limitations of RRAM-based CIM designs, which suffer from high read currents due to kΩ-

range cell resistance, necessitating large access transistors and restricting parallel compute rows to fewer than eight.

2.3.2 Macro Architecture

Each cell in the 64×64 FeFET crossbar encodes a binary weight: the low-VT (programmed) state represents logic '1' and the high-VT (erased) state represents logic '0'. Input features are translated into word-line voltages, and the cumulative drain current along each column constitutes the partial sum. A configurable pMOS pull-up transistor at the bitline end performs current-to-voltage conversion, feeding into voltage-mode 4-bit Flash ADCs. Each ADC consists of 15 voltage-mode sense amplifiers (VSAs) sharing a single ADC among eight columns. Weight precision is encoded by column grouping (1/2/4/8 bits), and input precision is encoded temporally across multiple compute cycles. The digital backend supports 1/8/16/64-row parallel compute configurations.

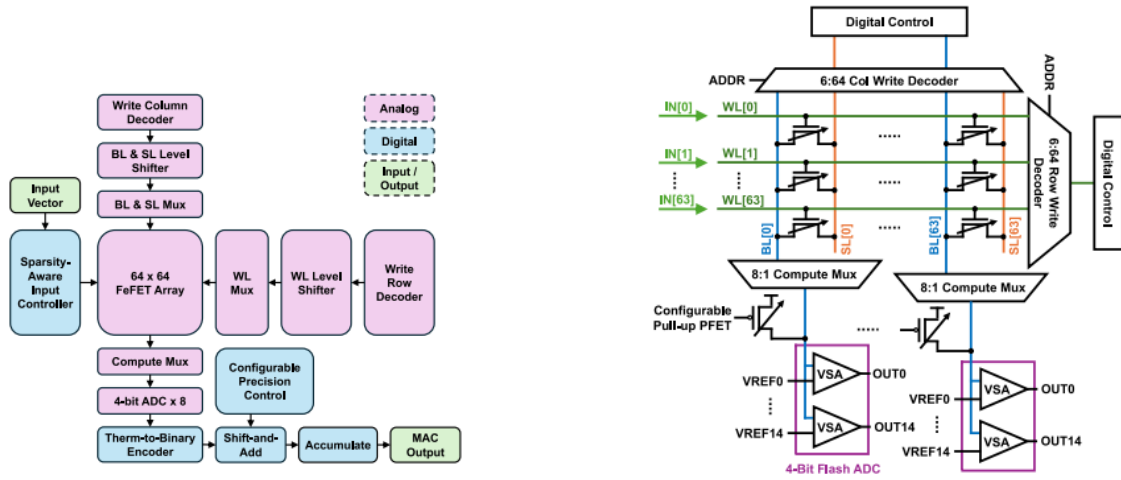


Figure 2.5: High-level block diagram and schematic of the 64×64 FeFET-CIM macro with peripheral circuits and 4-bit Flash ADC Figs. 2 and 3 from Garg et al. (IEEE SSCL, 2026)

2.3.3 Incremental Step Pulse Programming (ISPP) for Variation Reduction

A central contribution of this work is the application of an Incremental Step Pulse Programming (ISPP) scheme to mitigate the high I_{ON} current variation intrinsic to FeFET devices a variation arising from process non-uniformity, dipole relaxation, and charge trapping. The ISPP algorithm applies program pulses of increasing amplitude (2.5 V to 3.8 V in 50 mV steps, 5 μ s pulse width) until the device current at $V_{READ} = 1$ V reaches a target of 3 μ A. Prior to ISPP, the I_{ON} distribution had a mean of 4.85 μ A with a standard deviation of 819 nA. After ISPP, the mean was reduced to 3.27 μ A with a standard deviation of 377 nA a 54% reduction in spread that directly tightens the column current distribution and improves the sense margin of the ADC for distinguishing between partial sums. The erase operation is implemented using a positive bulk voltage (3.8 V, 10 μ s) rather than a negative gate pulse, avoiding the need for negative voltage generation circuits and ensuring chip-level compatibility.

2.3.4 DNN Inference Results and Benchmarking

The macro was evaluated on the VGG-8 model for the CIFAR-10 dataset using NeuroSim to process ADC error statistics. With 16-row parallel compute and 4-bit ADC resolution, the design achieved 85.2% inference accuracy, while 8-row parallel compute with 3-bit resolution yielded 89.1%, compared to a software baseline of 89.7%. The energy efficiency reached 346.6 TOPS/W for 1-bit $\times$ 1-bit MAC operations a 9.5 $\times$ improvement over the group's prior 40 nm RRAM-CIM macro. A simulated equivalent 64 $\times$ 64 RRAM design in 28 nm yielded only 78.2 TOPS/W, confirming that the efficiency gain is primarily attributable to the lower read current and leakage of FeFET cells, not merely the technology node scaling.

2.3.5 Significance

This work represents the state of the art in integrated FeFET-CIM macros, demonstrating the highest energy efficiency and compute density among FeFET-CIM crossbar designs with an integrated ADC at the time of publication. The use of ISPP, positive-voltage-only erase, and multi-row parallel compute together address the three primary engineering obstacles that had previously limited FeFET-CIM from practical deployment: variation, voltage compatibility, and parallelism.

2.4 An Efficient FeFET-Based CIM Macro Readout Inspired by the 3T Pixel Array Cell

2.4.1 Overview and Motivation

Laleni and Kämpfe (2024) from Fraunhofer IPMS and ETH Zurich proposed a novel FeFET-based IMC macro with a readout architecture inspired by the well-established three-transistor (3T) pixel array cell used in image sensor design [Laleni and Kämpfe, IEEE SMACD 2024]. The central observation motivating this design is that the analog-to-digital converter (ADC) in CIM accelerators typically dominates total analog power consumption, and that conventional ADC approaches often result in excessive area and power overhead. By borrowing the reset-sample-readout paradigm of the 3T pixel cell and reconfiguring it for array-level MAC readout rather than single-pixel readout, this work proposes a significantly more area- and power-efficient quantization path.

2.4.2 FeFET Cell and Variability Mitigation

The FeFET is used as a binary cell with two states: LVT (achieved by a +3.3 V positive gate pulse, polarizing the HfO₂ in one direction) and HVT (achieved by -4.5 V, reversing the polarization). Both states provide high data retention (greater than 10⁸ seconds at 125°C), supporting offline training and weight-stationary operation. To address LVT current variability, the design adopts the 1FeFET-1R (1F-1R) model with a source degeneration resistance of 5 M Ω placed in series with the FeFET, which limits drain current from the 1–10 μ A range down to approximately 100 nA. Rather than using a dedicated per-cell resistor (which would double the crossbar area), the design employs a segment-sharing strategy: a 16 $\times$ 16 FeFET array shares a single resistance element, and FETs sized at W $\times$ L

$= 500 \times 500 \text{ nm}^2$ are selected for optimal trade-off between cell area and readout error. The on-chip resistance is implemented as a MOSFET biased in saturation to avoid the large area penalty of a passive metal resistor.

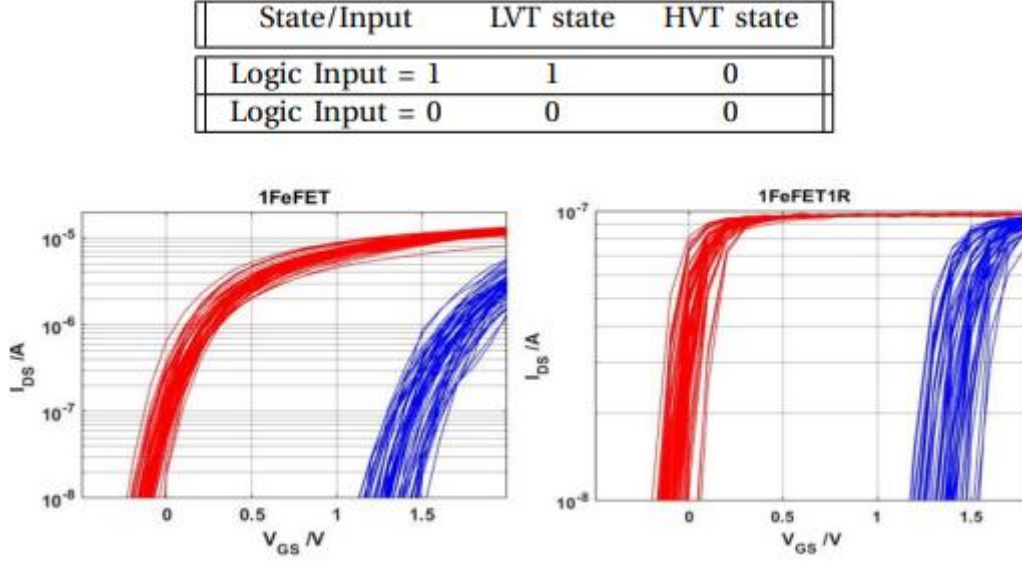


Figure 2.7: Measurement results comparing 1FeFET vs. 1FeFET-1R drain current characteristics, showing current limiting to $\sim 100 \text{ nA}$ with reduced spread — Fig. 2 from Lalen & Kämpfe (IEEE SMACD, 2024)

2.4.3 3T-Pixel-Inspired Readout Architecture

The proposed IMC macro consists of 16 column segments, each containing 16 FeFET rows, with 16 readout circuits shared via multiplexing. The readout operates in two distinct phases. In the first phase (reset), the bitline is precharged to VDD through a reset transistor (RST). In the second phase (evaluate), the word lines assert the FeFET inputs and the accumulated column current discharges the bitline capacitance, causing the bitline voltage to drop. A source follower (SF) transistor similar to the sensing transistor in the 3T pixel buffers the voltage change that occurs in the bitline during pixel measurement and segregates the bitline from the downstream comparator to prevent kickback.

The output is then latched by an inverter type comparator and a 1GHz synchronous counter counts time until the inverter latches thereby converting the discharge time into a digital word. The MAC output reference current which is constant thus linear has an output that is measured and used in post processing to obtain a linear digital version of the result.

The error between the ideal best-fit line and the simulated output is sufficiently small in all the activated FeFETs confirming that the time-to-digital conversion scheme is indeed linear.

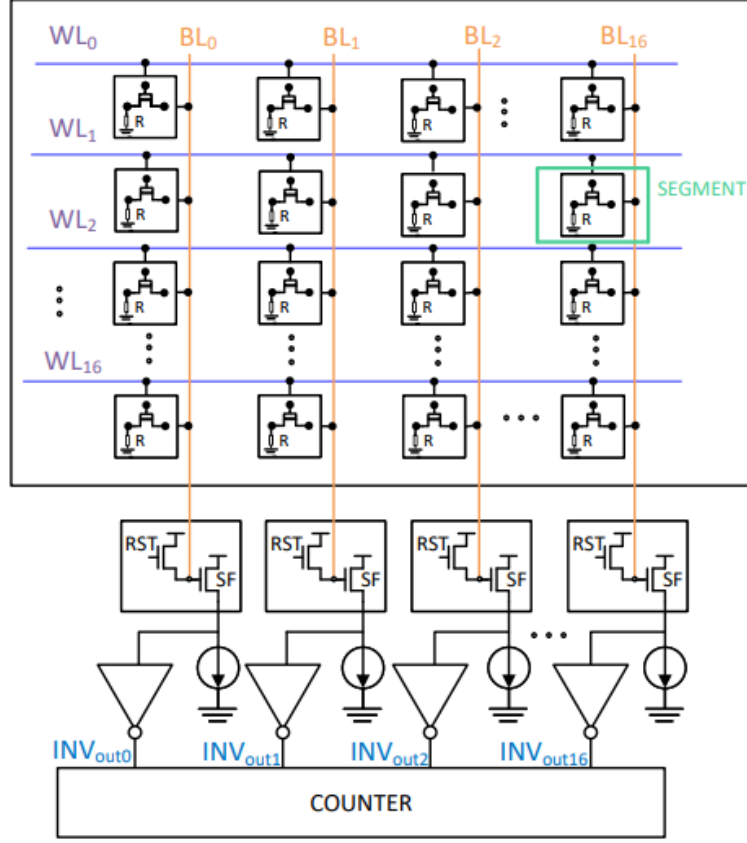


Figure 2.8: Full schematic of the proposed FeFET IMC macro showing the 16×16 segment array, RST/SF readout transistors, inverter comparator, and shared counter Fig. 3 from Lalenı & Kämpfe (IEEE SMACD, 2024)

2.4.4 Performance and Suitability for CNNs and SNNs

The supply voltage for this system is 0.8 V and it consumes a maximum average power of $8.18 \mu\text{W}$ for the read of one LVT FeFET. The ADC's sampling frequency is 8 MHz, which provides a Walden FoM of 63 fJ/conversion. This corresponds to an energy of 0.5 fJ per MAC operation in terms of CIM efficiency an attractive figure for edge AI applications. Increasing the design to 32 columns enhances the improvement to 0.25 fJ per operation. Thus the architecture is ideal for approximate computing frameworks and spiking neural networks (SNNs). As input sparsity implies that the full MAC range is hardly ever exercised, counter hardly counts to its maximum value.

2.4.5 Significance

The 3T-pixel-inspired readout offers a new concept which is different from that of Flash or SAR ADCs used in CIM macros. The design circumvents voltage-mode ADCs issues of precision mismatch and reference voltage tuning by transforming the MAC result into the time domain and quantizing discharge time with a digital counter. The area efficiency of the shared counter, along with its natural compatibility with sparse neural network workloads, makes this architecture a promising direction for future FeFET-CIM development particularly for SNN-based edge inference.

2.5 Comparative Summary and Research Gaps

In this chapter, the four manuscripts provide a timely overview of the FeFET based CIM technology evolution phase from proof-of-concept MAC demonstrations at 8×8 array level to fully integrated 4 kb macro prototypes with on-chip quantization. As can be seen in Table 2.1, a summary of other works is given below.

Parameter	De et al. (EDL 2022)	De et al. (TED 2022)	Garg et al. (SSCL 2026)	Laleni & Kämpfe (SMACD 2024)
Technology	28 nm HKMG	28 nm HKMG	28 nm HKMG	28 nm GF
Array Size	8×8	8×8	64×64	16×16 (sim.)
Variation Mitigation	1F-1T (transistor)	1F-1R (resistor)	ISPP scheme	1F-1R (segment)
ADC Type	Current-mode (on-wafer)	External	4-bit Flash (on-chip)	Counter-based (sim.)
Inference Accuracy	—	~97% (MNIST)	89.1% (CIFAR-10)	Linear MAC (sim.)
Energy Efficiency	—	—	346.6 TOPS/W	0.5 fJ/op
Retention	5×10^4 s (RT)	10^4 s @ 85°C	—	$>10^8$ s @ 125°C

Table 2.1: Comparative summary of reviewed FeFET-CIM works.

This review uncovers multiple research gaps and open challenges. To begin with, while each of the current-limiting transistors, external resistors, ISPP, has proven effective on its own, using a single method that can be scaled and can work with arrays larger than 64×64 without area penalty remains an open problems. Second, inference accuracy on complex datasets (CIFAR-10, ImageNet) degrades relative to the software baseline, and the accuracy-parallelism trade-off (fewer rows in parallel for better ADC sense margin) limits throughput. Third, most works report characterization at room temperature, and robustness under temperature variation critical for deployment requires further study. Fourth, write endurance of front-end-of-line FeFETs is limited to approximately 10^5 cycles, restricting online learning; BEOL-compatible and thin-film FeFET variants offer a potential path forward. Finally, the 3T-pixel-inspired readout architecture of Laleni and Kämpfe remains a simulation study and awaits experimental silicon validation.

2.6 Chapter Summary

This chapter has reviewed four pivotal works in the development of FeFET-based compute-in-memory macros for DNN inference. Beginning with the first experimental MAC demonstration on a 28 nm FeFET crossbar by De et al. (2022), the review traced the impact of current-variation mitigation via 1F-1T and 1F-1R structures, the system-level inference

accuracy achievable with resistor-shunted arrays (exceeding 97% on MNIST), and the most advanced integrated macro to date the 64×64 FeFET-CIM chip of Garg et al. (2026) achieving 346.6 TOPS/W energy efficiency. The novel 3T-pixel-inspired time-domain readout of Lalenı and Kämpfe offers a promising low-power quantization strategy. Together, these works establish the 28 nm GlobalFoundries HKMG FeFET platform as a mature and versatile substrate for CIM research, while identifying key challenges variation at scale, temperature robustness, endurance, and ADC efficiency that motivate the work presented in subsequent chapters of this thesis.

Chapter 3

Design and Implementation of Dual Port 8T SRAM-Based In-Memory Boolean Computation

This chapter presents the design and implementation of a Dual Port 8T SRAM-based architecture intended for in-memory Boolean computation. The paper focuses on executing logic gates for a set of fixed operands and the behaviour of the same at the circuit level. Details of the circuit architecture, functioning, simulation results and drawbacks are provided in order to assess the viability of the proposal.

3.1 Motivation for the Proposed Design

Boolean logic operations can be performed through in-memory computing methods based on SRAM. However, most existing methods store operands in separate memory cells and rely on extra sensing circuitry to read logic outputs. Circuit complexity is enhanced as well as the architecture overhead due to these demands. Hence, scalability and reliability can be constrained.

Motivated by these observations, a simple SRAM-based logic computation architecture can be designed to minimize circuit overheads. The aim is to assess whether basic Boolean logic functions can be executed on a conventional Dual Port 8T SRAM cell utilising its physically internal storage nodes and access paths. The logic evaluation can be enabled through a smaller and simpler circuit topology by removing shared read bitlines and dedicated sensing components.

The proposed architecture is evaluated in this work for the implementation of basic Boolean operations such as NAND, NOR, XOR and XNOR. The simulation focuses on functional correctness and circuit-level verification. This design is a targeted exploration of in-memory Boolean computing using a conventional Dual Port 8T SRAM cell. It exposes the behavior and limitations of in-memory computing against a conventional memory cell.

3.2 Dual Port 8T SRAM-Based Boolean Computation Architecture

In figure 4.1, architecture of the proposed Boolean processing architecture structure is shown. The Dual Port 8T SRAM cell is used for implementation in this design since it has internal storage nodes and access transistors to implement Boolean logic inside the SRAM cell. In contrast to many existing methods, this architecture eliminates the need for additional memory cells or shared read bitlines, enabling a more compact circuit realization.

In the proposed scheme, one operand is stored within the internal storage nodes of the SRAM cell, designated as QA and QB. The second operand is externally introduced through access transistors governed by the wordline, using complementary signals SPB and SNB. By suitably selecting the stored operand and the applied input signals, regulated discharge paths are established within the SRAM cell during the evaluation phase.

The bitline signals BL and BLB capture the logical interaction between the stored operand and the applied input. These signals are then processed through simple logic gates to produce Boolean outputs. As shown in Fig. 4.1, NOR and NAND outputs are derived through inverter-based sensing of BL and BLB respectively. Furthermore, XNOR and XOR operations are realized by applying AND and NAND logic to the BL and BLB signals. The logic gate combinations employed for output generation remain minimal and do not depend on skewed inverter-based sensing.

The proposed architecture facilitates the realization of fundamental Boolean operations using a single Dual Port 8T SRAM cell combined with standard logic gates. A thorough explanation of the logic operation mechanism and case-by-case behavior for NAND, NOR, XOR, and XNOR functions is provided in the following section.

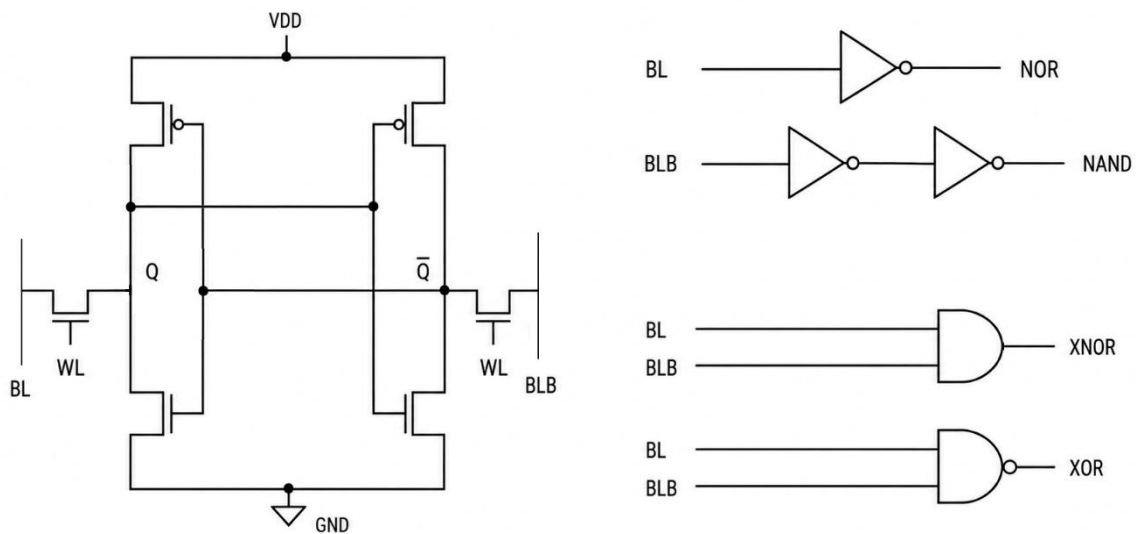


Figure 4.1: Dual Port 8T SRAM-Based In-Memory Boolean Computation Architecture.

3.3 Boolean Operation Mechanism

This section explains how Boolean logic operations are carried out using the proposed Dual Port 8T SRAM-based architecture. Operand A is stored at the internal storage node QA, while operand B is introduced through the access transistors via complementary control signals. The resulting voltage transitions on bitlines BL and BLB are utilized to evaluate the desired logic functions.

Figure Figure 4.2 provides reference Boolean outputs for all input combinations of the two operands A and B, which will come in handy for the NAND, NOR, XOR, and XNOR operations explained in the next subsections.

A	B	NAND	NOR	XNOR	XOR
0	0	1	1	1	0
0	1	1	0	0	1
1	0	1	0	0	1
1	1	0	0	1	0

Figure 4.2: Truth table for NAND, NOR, XNOR, and XOR operations.

3.3.1 NAND and NOR Operation

In the suggested architecture, operand A is kept in the internal storage node QA of the Dual Port 8T SRAM cell, while its complementary value is stored at QB. Both the bitlines BL and BLB are precharged to VDD prior to evaluation. NAND and NOR operations show that input signal SNB controls access transistor through which operand B is applied, as shown in Fig. 4.1. The control signals SPB and SNB are complements of each other to ensure conduction.

When both operands are logic '0' ($A=0, B=0$), the discharge path through the access transistor is activated such that BL is pulled low while BLB remains at the precharged high level. For input combinations '01' and '10', no effective discharge path is formed, causing both BL and BLB to remain at a high voltage level. When both operands are logic '1' ($A=1, B=1$), BL remains high while BLB is discharged to a low voltage level.

Based on this bitline behavior, the NOR output is obtained by inverting the voltage level on BL, while the NAND output is taken directly from BLB through a buffer stage. The resulting logic levels correspond precisely to the expected NAND and NOR truth table shown in Fig. 4.2. This operational mechanism is depicted in Fig. 4.3 and demonstrates the realization of NAND and NOR functions within a single Dual Port 8T SRAM cell, without requiring skewed inverter sensing or shared read bitlines.

Q_A=A	SN_B=B	BL	BLB
0	0	↓ 0	1 ↑
0	1	1 ↑	1 ↑
1	0	1 ↑	1 ↑
1	1	1 ↑	↓ 0

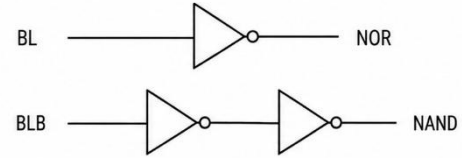


Figure 4.3: Bitline behavior and logic extraction for NAND and NOR operations.

3.3.2 XOR and XNOR Operation

The proposed Dual Port 8T SRAM-based architecture additionally supports XOR and XNOR operations by appropriately combining the bitline signals BL and BLB produced during evaluation. As with the NAND and NOR case, operand A is stored at internal node QA, while operand B is applied through the access transistors using complementary control signals SPB and SNB. The corresponding logic-level behavior of BL and BLB for various input combinations is summarized in Fig. 4.2 and illustrated in Fig. 4.4.

In the course of operation, the BL and BLB reflect the logical relationship stored and applied operands. In case of input combinations $A=B$, since the bitlines assume complementary voltage levels, their logical “AND” is a high output. The output of the BL and BLB logical AND is low when the operands are different ($A \neq B$). The XNOR output is produced from the AND of BL and BLB signals.

The same BL and BLB signals are NANDed together to create an XOR output. The logical combination will produce a high output at the XOR when the two operands differ and a low when they are the same as shown in the truth table of Fig. 4.2. Using simple logic gates for output generation eliminates the requirement for elaborate sensing circuitry or detection via skewed inverters.

The combination of two bitline signals through logic levels can achieve XOR and XNOR functions in a Single Dual Port 8T SRAM Cell. The circuit-level simulations presented in the next subsection confirm the functional correctness of these operations.

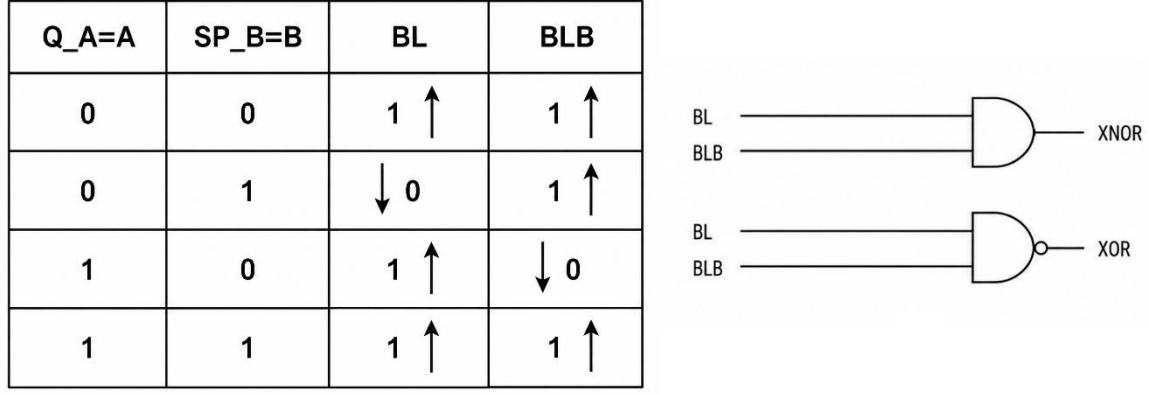


Figure 4.4: Bitline behavior and logic extraction for XNOR and XOR operations.

3.4 Transistor-Level Implementation and Simulation Results

The Dual Port 8T SRAM-based Boolean computation architecture was realized at the transistor level employing standard CMOS devices. The entire circuit diagram of SRAM cell, access transistors and output logic gates is shown in fig. 4.5. Cadence Virtuoso was used for the simulation and implementation of the circuit to check functional correctness.

The transient runs were performed to check the circuitry behavior for all the combinations of inputs A and B. Prior to every assess, the bitlines BL and BLB where precharged to VDD followed by activating the wordline and applying the control signals. As stated in Section 4.3, the voltage changes on BL and BLB were monitored, and logic outputs were generated using buffering, inversion, AND and NAND logic blocks.

Simulation waveforms for NAND, NOR, XOR, and XNOR operations are presented in Figs. 4.6–4.13 for input combinations '00', '01', '10', and '11'. The observed outputs align with the expected logic levels summarized in Table 4.2, confirming the correct functional operation of the proposed architecture.

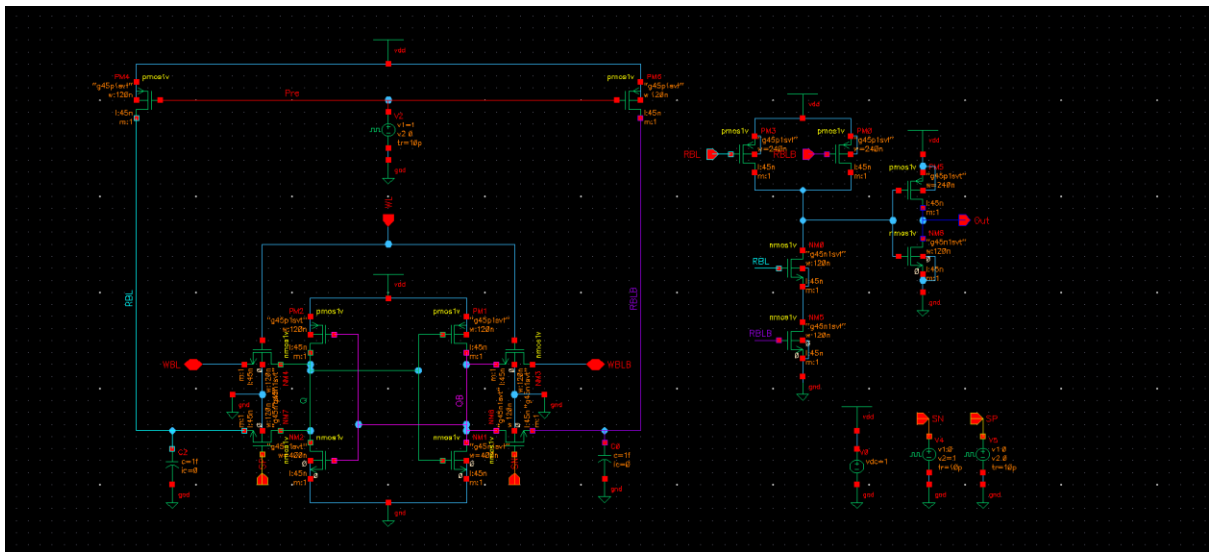


Figure 4.5: Transistor-level schematic of the proposed 8T SRAM-based Boolean computation architecture.

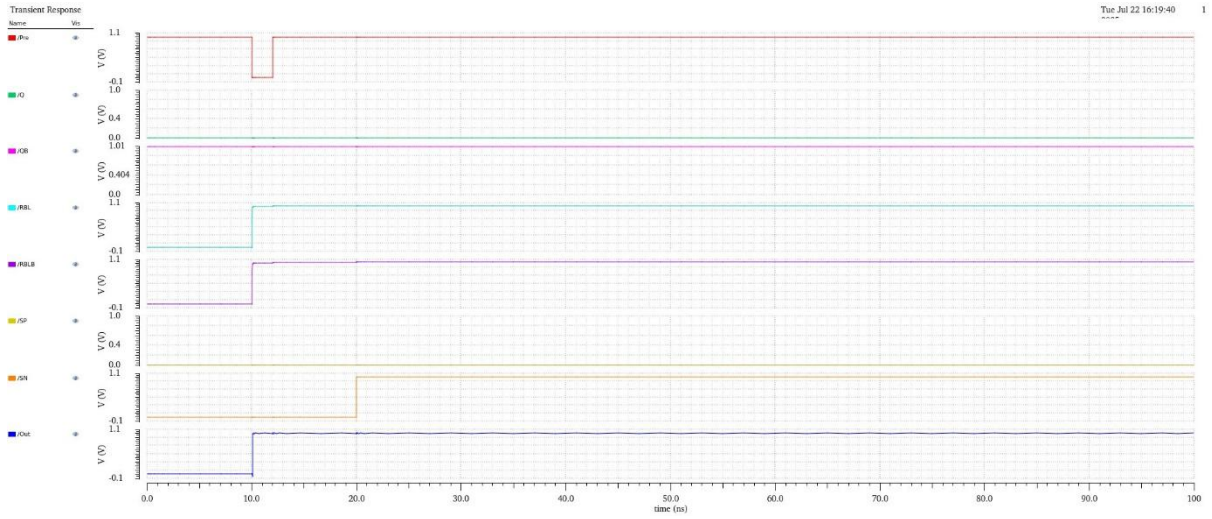


Figure4.6: $Q=0$, $QB=1$, $SP=0$ and $SN=1$

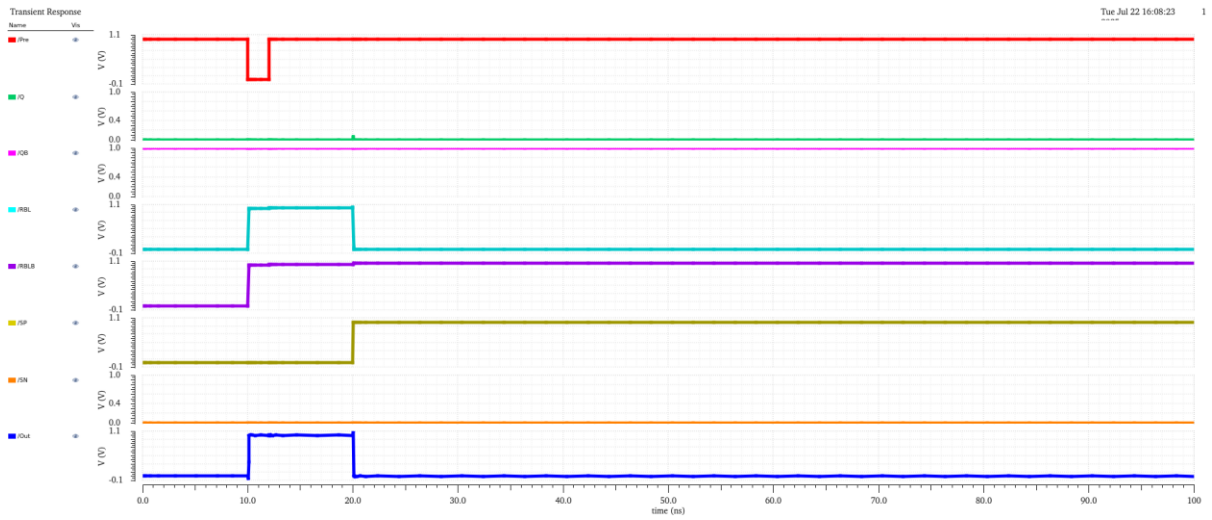


Figure4.7: $Q=0$, $QB=1$, $SP=1$ and $SN=0$

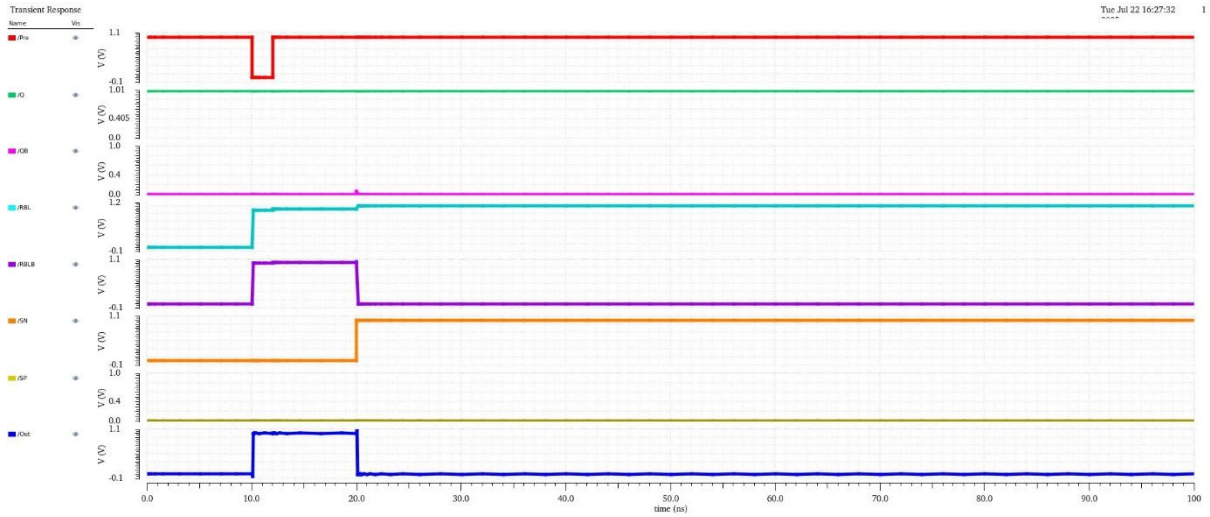


Figure4.8: $Q=1$, $QB=0$, $SP=0$ and $SN=1$

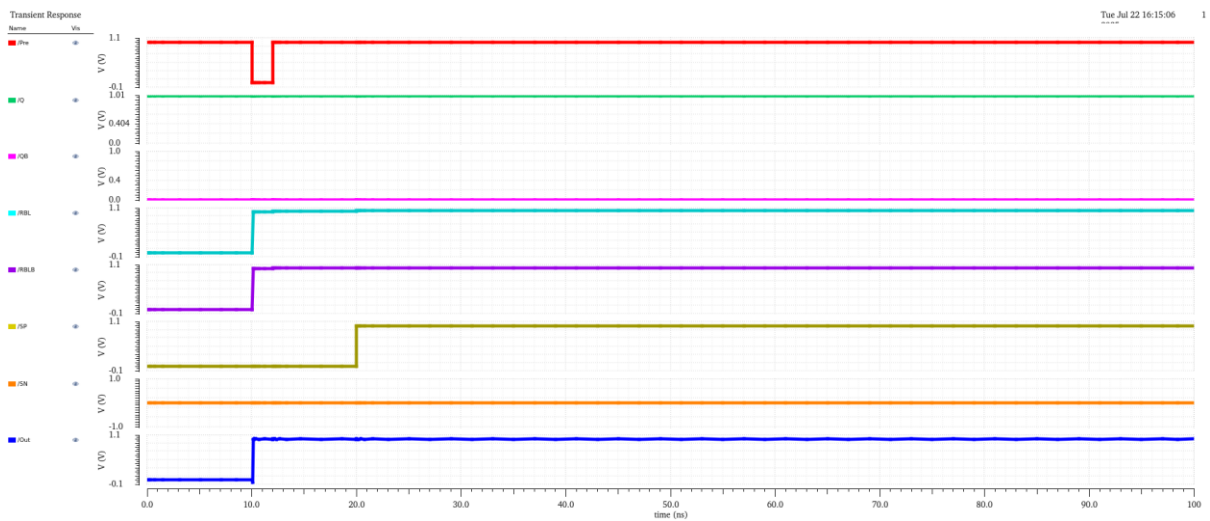


Figure4.9: $Q=1$, $QB=1$, $SP=1$ and $SN=0$

- RBL gives OR , when passed through inverter gives NOR ($Q=A$, $SN=B$)
- $RBLB$ gives $NAND$, when passed through inverter gives AND ($Q=A$, $SN=B$)
- Output of RBL and $RBLB$ is passed through and gate it gives $XNOR$ ($Q=A$, $SP=B$)
- $XNOR$ result passed through the inverter gives XOR

Chapter 5

Conclusion

An 8×8 FeFET-based analog Compute-in-Memory MAC architecture was successfully designed and simulated in Cadence Virtuoso using GPDK 45 nm technology, with VDD = 1.0 V and 8-bit input/weight resolution. The key contributions and findings are summarized below:

- **FeFET Device Model:** A BSIM Verilog-A compact model captures ferroelectric P–E hysteresis and Vth bistability (memory window ≈ 1.0 – 1.5 V), instantiated as a schematic symbol in Cadence Virtuoso for all circuit-level simulations.
- **Write Protocol:** Each FeFET cell is programmed via 5 bipolar wake-up cycles (± 5 V, 50 ns each) to condition the ferroelectric domains, followed by a single state-setting pulse (+5 V for LVT/'1', -5 V for HVT/'0'), totalling 550 ns per cell.
- **Non-Disturbing Read:** A 100 mV drain pulse preserves ferroelectric polarization throughout inference, ensuring non-destructive and reliable weight readout.
- **4-Cycle PWM MAC Operation:** The 8-bit input is segmented into four 2-bit groups (B_3 – B_0), each encoded as proportional PWM wordline pulses and processed in a dedicated computation cycle, enabling full 8-bit MAC precision across four pipelined cycles.
- **Hierarchical Analog Accumulation:** A 1C-2C-4C capacitive shift-and-add network combines four intermediate partial products (V_1 – V_4) into two digitization-ready signals (Y_0 , Y_1) per cycle, achieving a 50% reduction in ADC count ($32 \rightarrow 16$) and directly lowering area and power overhead.
- **Digital Post-Processing:** Four cycle-level partial sums (P_0 – P_3) are digitally shifted and accumulated to recover the full 8-bit MAC result with high arithmetic precision.
- **Non-Volatility Advantage:** FeFET's inherent non-volatility eliminates refresh power, enables instant-on operation, and ensures reliable weight retention at zero standby current — key advantages over SRAM- or DRAM-based CIM architectures.
- **Measured Performance:**
 - Average Power: **98 μ W**
 - Latency: **16 ns** per cycle
 - Throughput: **2.13 GOPS**
 - Energy Efficiency: **8.59 TOPS/W**

- **Scalability and Impact:** The proposed architecture demonstrates that ferroelectric non-volatile memory can be effectively leveraged for energy-efficient, accurate analog in-memory computation, providing a scalable and power-efficient path to 8-bit neural network inference at the edge.

Chapter 4

FeFET-Based 8-Bit CIM Array MAC Operations

4.1 Introduction on FeFET based MAC

The increasing demand of data-hungry machine-learning workloads like CNNs and transformer models is exposing a fundamental bottleneck in conventional von Neumann computing architecture, namely, the continuous transfer of data between memory and processing components that are located far apart from each other, also known as the memory-wall problem. This limited bandwidth paradigm creates energy misallocation, which is highly expensive for edge inference applications where power budgets are highly constrained.

Compute-in-Memory (CIM) architectures have emerged as a promising solution that embeds multiply-and-accumulate (MAC) computation either in memory array or around the memory array to avoid data movement overheads. Owing to their non-volatile programmable multi-state feature, ferroelectric materials combined with their gate controlled transconductance behavior, offer an exciting synergy in the form of Ferroelectric Field-Effect Transistors (FeFETs) greatly facilitating their use in NVM technologies. It further facilitates in situ analog computation for very low read power.

This chapter presents a detailed architectural description, operational protocol, and MAC computation methodology for the proposed 8×8 FeFET-based analog CIM array. The architecture is described in the context of GlobalFoundries' 28 nm High-K Metal Gate (HKMG) process technology, wherein 8 nm silicon-doped hafnium oxide (Si:HfO₂) serves as the ferroelectric gate dielectric. The chapter elaborates the device physics, binary weight programming scheme, read pulse conditions, and a full cycle-by-cycle 8-bit MAC operation employing pulse-width modulation (PWM) of input activations and a capacitive shift-and-add analog accumulation strategy.

The simulator used for this work is Cadence Virtuoso with GPDK 45 nm PDK. The FeFET device symbol and behavioral model were realized using a BSIM-based SPICE model written in Verilog-A, whose behavioral description faithfully captures ferroelectric polarization

switching, remanent polarization, and the resulting threshold-voltage hysteresis. This symbol was subsequently instantiated in all schematic-level simulations.

4.2 FeFET Device: Technology, Structure, and Fundamentals

4.2.1 What is a FeFET?

A Ferroelectric Field-Effect Transistor (FeFET) is a non-volatile transistor that uses a ferroelectric material as its gate dielectric to store data by retaining programmable polarization states. Structurally, a FeFET is a metal-oxide-semiconductor field-effect transistor (MOSFET) in which the conventional gate dielectric (SiO_2 or high- κ oxide) is replaced either fully or in part by a ferroelectric layer. In contemporary CMOS-compatible implementations, the ferroelectric material is hafnium oxide doped with zirconium ($\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$, or HZO), which exhibits robust ferroelectricity in films as thin as 5–10 nm.

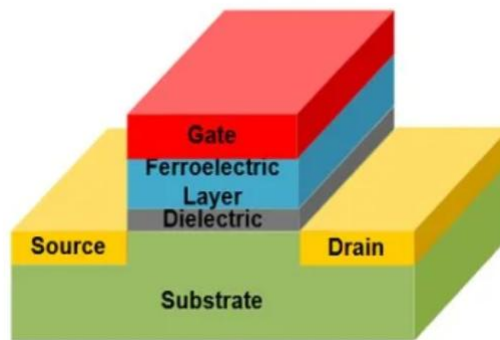


Figure 4.1: Structure of FeFET

The key property of the ferroelectric layer is its spontaneous electric polarization a bistable alignment of electric dipoles that can be switched by an external electric field and that persists at zero bias due to the remanent polarization (P_r). This bistability directly translates into two distinct, non-volatile threshold-voltage states in the FeFET, making it an ideal weight-storage primitive for neural network hardware.

4.2.2 Technology Platform

The FeFET devices studied in this work utilize 300 mm wafers from GlobalFoundries' 28 nm HKMG technology platform. An approximately 1 nm silicon dioxide is interfacial layer that lies between the ferroelectric and semiconductor channel of the gate stack which integrates an 8 nm ferroelectric layer silicon-doped hafnium oxide. This gate stack can be processed using standard CMOS BEOL processing and thus can be integrated with VLSI designs without the addition of exotic processing steps.

For performing simulation Cadence Virtuoso with GPDK 45 nm PDK is used as design environment. The FeFET device used in this work is modeled with a BSIM compact model that extends with the Verilog-A database of the ferroelectric polarization dynamics. Designed based on experimental observations, this model matches the polarization–electric field (P–E) hysteresis loop, E_c , and P_r , which are essential for the ΔV_{th} that modulates the channel of the underlying MOSFET. Cadence Virtuoso was used to create the resulting FeFET symbol which was used in all schematic simulations.

4.2.3 Ferroelectric Polarization and Threshold Voltage Modulation

The fundamental operating principle of the FeFET exploits the non-volatile, electrically switchable spontaneous polarization of the ferroelectric HfO_2 layer. The ferroelectric material exhibits a characteristic polarization–electric field (P–E) hysteresis loop due to bistable domain switching. The two key parameters of this loop are:

- Remanent Polarization (P_r): The residual polarization that remains after the applied field is removed. A large P_r ensures wide, stable threshold-voltage states.
- Coercive Field (E_c): The minimum field required to switch the polarization direction. It defines the minimum programming voltage.

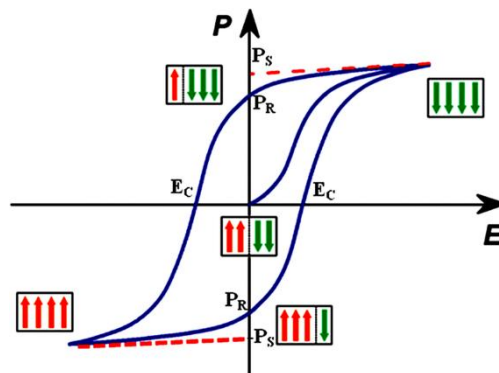


Figure 4.2: Hysteresis (P-E) curve in ferroelectric materials

When a sufficiently large positive gate voltage (exceeding $+E_c$) is applied, the ferroelectric dipoles align in a direction that increases the surface charge density at the ferroelectric–semiconductor interface, effectively lowering the threshold voltage (V_{th}) to the Low-Threshold-Voltage (LVT) state. Conversely, a sufficiently large negative gate voltage reverses the dipole orientation, reducing interface charge and elevating V_{th} to the High-Threshold-Voltage (HVT) state. Both states are stable at zero gate bias, making the FeFET inherently non-volatile.

The two binary storage states are defined as:

- Low-Threshold-Voltage (LVT) State Logic ‘1’: The device exhibits higher channel conductance (G_{th}) and lower threshold voltage. Set by a positive gate pulse.
- High-Threshold-Voltage (HVT) State Logic ‘0’: The device exhibits lower channel conductance and higher threshold voltage. Set by a negative gate pulse.

The threshold voltage window (Memory Window, MW) is the primary figure of merit:

$$MW = V_{th,HVT} - V_{th,LVT}$$

The FeFET gate stack of the proposed architecture is designed to have MW ranging from ≈ 1.0 – 1.5 V, which is adequate to provide a good noise margin for the analog bitline discharge operation.

4.3 Key FeFET Array Characteristics

4.3.1 Non-Volatility

The array based on FeFET is non-volatile in nature. When the power is off, stored weights remain in the ferroelectric polarization state, enabling instant-on operation and alleviating weight-reload overhead. It is especially important for duty-cycled edge inference devices that are often powered down between inference activities.

4.3.2 Compact Cell Structure

Each FeFET cell uses one transistor architecture to save a single weight bit. The compact design allows for highly dense integration of the weight-storage elements in 8×8 CIM arrays and is area-efficient in comparison to multi-transistors.

4.3.3 Programming Voltage Requirements

FeFET programming requires ± 5 V bipolar pulses for polarization switching. Dedicated high-voltage generation and level-shifting circuits are therefore required in the programming path. These circuits are isolated from the low-voltage (1.0 V VDD) analog compute path during inference, ensuring that the programming overhead does not impact inference energy.

4.3.4 Read Disturb Considerations

The read wordline voltage (100 mV) is selected well below the coercive voltage of the ferroelectric layer. This ensures stable long-term operation with negligible threshold-voltage drift during inference, even under repeated computation cycles.

4.3.5 Endurance

FeFET endurance is limited by ferroelectric fatigue caused by repeated polarization switching. However, for edge inference applications where weight updates occur infrequently (weights are programmed once and read many times), the endurance characteristics are sufficient for long-term operation. The dominant operating mode is read (inference), not write (programming).

4.4 FeFET Device Symbol and BSIM Model in Cadence Virtuoso

To enable circuit-level simulation in Cadence Virtuoso, a FeFET device symbol was created using a Verilog-A compact model based on the BSIM framework. The Verilog-A code realises the ferroelectric polarization dynamics, such as the P–E hysteresis, coercive field switching, and mapping of polarization state to threshold-voltage shift. The BSIM MOSFET core takes care of all the standard transistor characteristics. The contribution from the ferroelectric layer is added as a modulation onto the gate effective voltage.

The design-based symbol was imported into the Cadence Virtuoso schematic environment and connected as an ordinary four-terminal device (Gate, Drain, Source, Bulk). The physical accuracy of the ferroelectric switching behavior of our devices in all operating conditions has been ensured using this symbol for all array simulations including wake-up cycling, binary weight programming, read sweeps, and MAC computation.

The Verilog-A model based on BSIM can reproduce the following essential behaviors required for CIM operation, such as threshold-voltage bistability between LVT and HVT states, hysteretic switching as a function of gate pulse amplitude and width, remanent polarization retention at zero bias, and differential drain current response enabling analog bitline discharge.

4.5 Weight Programming Protocol (WRITE Operation)

To accomplish binary weights, it is important to note that FeFET cells require voltages that can be well above VDD, typically ± 5 V. Additionally, newly manufactured (or previously inactive) FeFET devices exhibit imprint: charge that accumulates at the ferroelectric–dielectric interface pushes polarization to one state, resulting in asymmetric switching due to the imprint and a shift in V_{th} distributions. Before a reliable binary programming can be performed, a forced wake-up cycling step eliminates this imprint.

As a result, the total programming sequence for each FeFET cell in the proposed 8×8 array appears to consist of two components, including (1) wake-up conditioning and (2) state-setting LVT ('1') or HVT ('0') writing. Dedicated peripheral programming circuits perform both stages, while isolated from the analog compute path during inference.

4.5.1 Stage 1: Wake-Up Cycling

The process of wake-up cycling involves applying a bipolar voltage pulse to a FeFET to wake it up before its first use as a memory element. The physical intent is to disintegrate pinned polarization domains and uniformly re-initiate the ferroelectric grain ensemble within the HZO film to maximize $2P_r$ and the generated threshold-voltage window.

In the proposed architecture, each FeFET undergoes exactly five wake-up cycles immediately before weight programming. Each individual cycle consists of two sequential pulses:

- A positive pulse of amplitude $+5$ V and duration 50 ns, which drives the polarization into the HVT orientation.
- A negative pulse of amplitude -5 V and duration 50 ns, which drives the polarization back into the LVT orientation.

The five consecutive bipolar cycles of ± 5 V / 50 ns serve to activate pinned ferroelectric domains, broaden the hysteresis loop and stabilize remanent polarization towards the maximum value. The fifth repetition of the distribution is narrow and symmetric enough to ensure reliable binary weight storage with sufficient read margin after five complete cycles.

While cycling to wake up, some rows and columns of the array (de-selected) are held at half-select voltages (± 2.5 V) using a normal inhibit scheme.

Consequently, only the target FeFET cell is subjected to a full ± 5 V across its ferroelectric gate, while unselected adjacent cells experience at most ± 2.5 V below their coercive voltage, avoiding inadvertent polarization switching.

4.5.2 Stage 2: Writing Logic ‘1’ (LVT State)

Following the five wake-up cycles, the FeFET is in a conditioned but indeterminate state. To write a logic ‘1’ weight corresponding to the LVT state a single final programming pulse of amplitude +5 V and duration 50 ns is applied to the gate of the target FeFET. Source, drain, and bulk are maintained at 0 V.

The large positive gate voltage generates an electric field across the HZO ferroelectric layer that exceeds the positive coercive field ($+E_c$). This field aligns the polarization dipoles in the direction that induces a net negative bound charge at the ferroelectric–semiconductor interface, which produces an accumulation of mobile electrons in the n-type channel. The effective threshold voltage is thereby reduced to $V_{th,LVT}$ (typically 0.2–0.5 V). The device conducts significantly at typical read wordline voltages (0.6–0.8 V) and is classified as storing a logic ‘1’.

4.5.3 Stage 2: Writing Logic ‘0’ (HVT State)

To write a logic ‘0’ weight corresponding to the HVT state a single final programming pulse of amplitude –5 V and duration 50 ns is applied to the gate. The large negative gate voltage generates a field that exceeds the negative coercive field ($-E_c$), aligning the ferroelectric polarization in the direction that induces net positive bound charge at the ferroelectric–semiconductor interface. This depletes the n-channel, raising the effective threshold voltage to $V_{th,HVT}$ (typically 1.2–2.0 V). The device conducts negligibly at the read wordline voltage, correctly encoding a weight of ‘0’.

4.5.4 Complete Programming Sequence Summary

The complete programming sequence for any single FeFET weight cell in the 8×8 array is an eleven-pulse protocol totaling 550 ns per cell:

Pulse	Pulse Type	Amplitude	Duration	Purpose
1–10	Wake-up ($\times 5$ bipolar)	± 5 V	50 ns each	Condition ferroelectric domains
11	State-setting (+)	+5 V	50 ns	Write Logic ‘1’ (LVT)
11	State-setting (–)	–5 V	50 ns	Write Logic ‘0’ (HVT)

Table 4.1: FeFET Binary Weight Programming Protocol Summary

Since inference is the primary operating mode and programming occurs only during weight updates, this 550 ns per-cell overhead is acceptable for the targeted edge inference use case.

4.6 READ Operation and Sensing Mechanism

The READ operation must be performed in a non-disturbing manner the applied gate voltage must not be sufficiently large to switch the polarization state of the ferroelectric layer, thereby preserving the stored weight indefinitely. In the proposed architecture, the read wordline voltage is selected well below the coercive voltage of the ferroelectric layer.

A small read pulse of 100 mV is applied at the drain terminal, while the gate terminal receives the wordline pulse. The source and bulk terminals are maintained at 0 V throughout the READ operation. The non-disturbing read pulse allows the sense circuitry to differentiate between the two weight states based on the channel current response:

- LVT Cell (weight ‘1’): Lower threshold voltage means the device turns on at smaller gate voltage. For a given wordline bias with $V_D = 100$ mV, the drain current (I_D) is comparatively higher, causing a larger bitline voltage discharge.
- HVT Cell (weight ‘0’): Elevated threshold voltage results in reduced channel conductance. For the same gate bias and $V_D = 100$ mV, I_D is substantially smaller, resulting in negligible bitline discharge.

The effective READ mechanism during MAC computation of the CIM array utilizes PWM wordline pulses of 100 mV amplitude. The choice of bitline discharge during effective READ is done based on both the stored weight and input pulse width. This effective READ has been explained in detail in Section 4.6.

4.7 8×8 FeFET Array Architecture

4.7.1 Array Organization

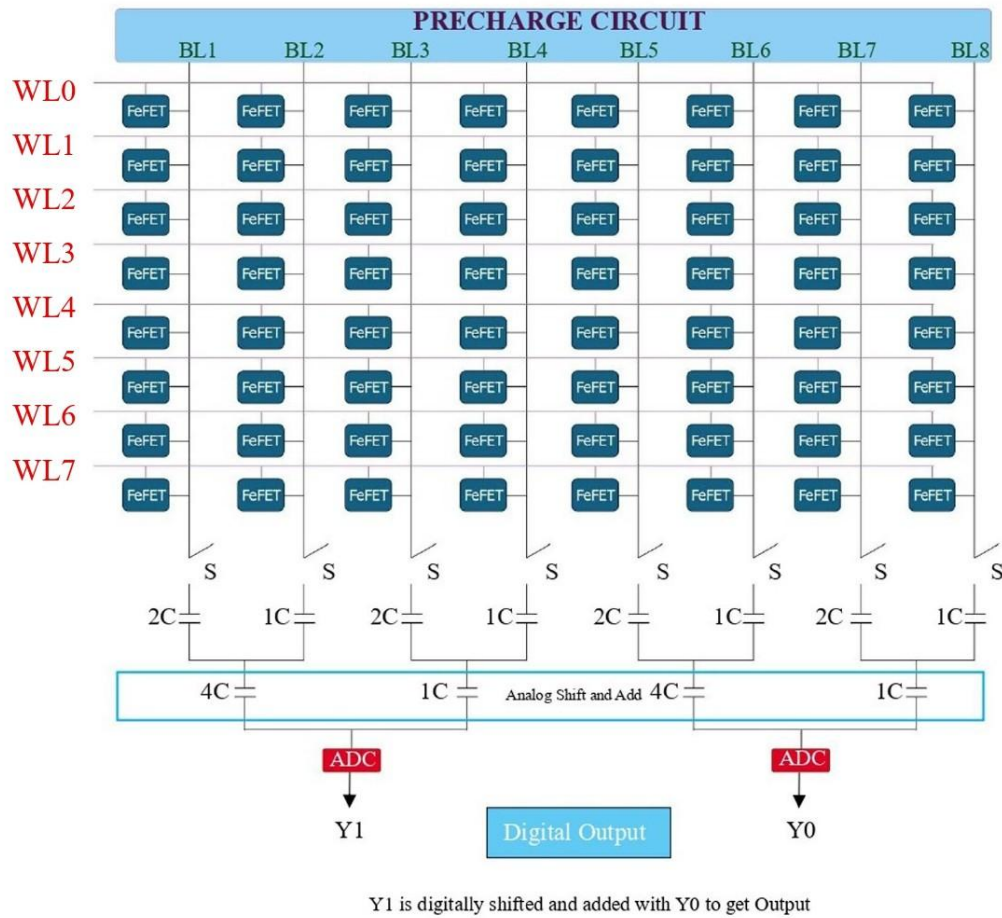


Figure 4.3: Proposed 8×8 FeFET Array Architecture

The proposed array of FeFET CIM is organized by 8-row and 8-column which gives a total of 64 FeFET cells. Each cell contains one bit of binary weight. The array performs time-multiplexed multi-cycle operation with analog accumulation to enable 8-bit MAC computation. The specifics of the design are below.

Parameter	Specification
Simulator	Cadence Virtuoso
PDK	GPDK 45 nm
Device	FeFET (BSIM + Verilog-A model)
Supply Voltage (VDD)	1.0 V
Array Size	8 × 8 FeFET cells (64 cells total)

Parameter	Specification
Bit Width	8-bit input & 8-bit weight
Programming Voltage	± 5 V bipolar pulses
Read WL Voltage	100 mV (non-disturbing)
Technology Node	GlobalFoundries 28 nm HKMG
Ferroelectric Material	Si:HfO ₂ (8 nm, Hf _{0.5} Zr _{0.5} O ₂)

Table 4.2: Design Specifications of the Proposed FeFET CIM Array

The array organization is as follows:

- Each row is linked to one WL (wordline) (8 total). During a computing cycle, a pulse from the PWM is applied by encoding a row corresponding to WL. One cycle can simultaneously process eight independent input activations owing to the presence of eight rows.
- Columns (8 total): Each column corresponds to one bit of the filter weight vector. The bitlines (BLs) of each column accumulate the partial discharge results from all active rows, implementing a charge-mode dot product.
- FeFET Cells: Each cell at the intersection of a row and column stores one binary weight bit. The cell's weight (LVT='1' or HVT='0') determines whether, and to what degree, the cell contributes to the bitline discharge during a given computation cycle.

4.7.2 Weight Mapping Scheme

The 8-bit filter weight for each row is mapped directly across the 8 FeFET cells of that row, one bit per cell, from the most significant bit (W7) to the least significant bit (W0). The MSB (W7) occupies the cell in column 7, and the LSB (W0) occupies the cell in column 0. This bit-serial column mapping allows the capacitive accumulation network in each column to implement a weighted contribution based on positional significance.

4.7.3 2-Bit Weight Grouping and Capacitive Network

To implement an efficient analog 8-bit MAC operation, the 8-bit weight stored across the 8 cells of a row is subdivided into four 2-bit groups, corresponding to the input segmentation described in Section 4.6:

- Group G3 [W7, W6]: Most significant 2-bit weight segment.
- Group G2 [W5, W4]: Second weight segment.
- Group G1 [W3, W2]: Third weight segment.
- Group G0 [W1, W0]: Least significant 2-bit weight segment.

Within each 2-bit weight group, the most significant weight bit is connected to a capacitor of value $2C$, and the least significant weight bit is connected to a capacitor of value $1C$. This $1C$ - $2C$ capacitive network implements an analog binary weighting within each group, enabling

partial MAC output voltage contributions to be naturally summed in the charge domain during the charge-sharing phase.

Furthermore, an inter-group capacitive scaling network using a 4C capacitor connects the outputs of adjacent groups in a hierarchical shift-and-add structure. This allows the analog shift-and-add operation to combine partial MAC outputs across all four groups without requiring a 64C capacitor (which would be needed for a 6-bit left shift), thereby constraining the capacitive area overhead while maintaining arithmetic accuracy.

4.8 8-Bit MAC Operation: Four-Cycle Protocol

4.8.1 Input Encoding via Pulse-Width Modulation (PWM)

The 8-bit input feature vector for each row is encoded into a series of PWM pulses applied to the respective wordline. Each PWM pulse has a fixed amplitude of 100 mV identical to the non-disturbing read pulse ensuring that the polarization state of the FeFET cells is not altered during computation. The pulse width is varied in proportion to the numerical value of the 2-bit input segment being processed in each cycle.

The 8-bit input is partitioned into four sequential 2-bit segments. Consider the example input $I = 11001001_2$:

$$I = 11001001_2$$

$$B_3 = 11_2 = 3 \text{ (bits [7:6], MSB)}$$

$$B_2 = 00_2 = 0 \text{ (bits [5:4])}$$

$$B_1 = 10_2 = 2 \text{ (bits [3:2])}$$

$$B_0 = 01_2 = 1 \text{ (bits [1:0], LSB)}$$

Each 2-bit segment B_k ($k = 0, 1, 2, 3$) has a decimal value ranging from 0 to 3. A tapped delay-line chain of voltage-controlled delay buffers (VCBs) generates a PWM pulse whose width is proportional to this decimal value.

Each 2-bit section is converted into relative pulse width of 100mv and pulse width is input a WLs

4.8.2 Four Computation Cycles

The complete 8-bit MAC operation for the 8×8 array is executed over four sequential computation cycles, each corresponding to one 2-bit input segment:

- Cycle 1: PWM pulse corresponding to segment B_0 (bits [1:0]) LSB segment.
- Cycle 2: PWM pulse corresponding to segment B_1 (bits [3:2]).

- Cycle 3: PWM pulse corresponding to segment B₂ (bits [5:4]).
- Cycle 4: PWM pulse corresponding to segment B₃ (bits [7:6]) MSB segment.

Within each of the four cycles, the full MAC computation sequence proceeds through four distinct phases: pre-charge, discharge, charge-sharing, and output digitization. These phases are described in detail in Section 4.7.

4.8.3 Parallel Row Activation

One of the characteristics of the 8x8 array is that all 8 wordlines are fired at once during a compute cycle. The PWM pulses for the same 2-bit section of each of the 8 input feature vectors are produced in parallel and simultaneously applied to their corresponding WLs. As a result, the array can perform 8 partial dot-products simultaneously during each cycle, and the bitline of each column gathers discharge contributions from all 8 activated rows.

Each row is assigned a unique input value (and thus a unique PWM pulse width), and since each cell holds a different weight bit, the bitline voltage at the end of the discharge phase is effectively an analog sum of the product of each row's input-encoded pulse width and its weight bit value. The principal analog dot product operation.

4.9 Detailed Phase-by-Phase Operation

4.9.1 Phase 1: Pre-Charge

At the start of each computation, all bitlines BLs in the 8x8 array are charged at the supply voltage VDD. This determines a known initial state for the analogue accumulation. The precharge transistors (one per column) are enabled for a short time to fully charge all the column bitlines. This is then followed by the disabling of the precharge transistors, leaving the bitlines floating (preparation for discharge).

During the precharge phase, the wordlines of all rows are maintained at 0 V, ensuring no FeFET cell conducts and preventing any bitline discharge before the computation begins.

4.9.2 Phase 2: Discharge (PWM-Driven, FeFET-Dependent)

During the phase of discharging, the pulses of the current cycle that are encoded in PWM are applied to all 8 wordlines concurrently. Each wordline receives a 100 mV pulse whose duration TWL encodes the 2-bit input segment for that row. The LVT state of a FeFET cell means that the bitlines will start discharging through it since its threshold is lower and this produces a 100mV WL pulse which will turn this cell on.

The physics of the bitline discharge are governed by the FeFET transconductance and the bitline parasitic capacitance CBL. For a single FeFET cell in the LVT state:

$$\Delta V_{bl} = \frac{W_d C_{ox} V_{dsat} (V_{wl} - V_{th}) w T_{wl}}{C_{bl}}$$

Since $w = 1$ for LVT (significant overdrive) and $w = 0$ for HVT (negligible overdrive with $V_{WL} = 100$ mV below coercive voltage), the bitline discharge is selectively enabled only for cells storing weight ‘1’. The total bitline voltage change after one cycle’s discharge phase thus reflects the analog dot-product of the input PWM pulse widths with the stored binary weight column.

4.9.3 Phase 3: Charge Sharing (1C-2C Capacitive Network)

Following the discharge phase, switch S1 is activated to initiate charge redistribution among the capacitors in the 1C-2C capacitive network. This phase implements the analog shift-and-add accumulation that aggregates partial MAC contributions from the four 2-bit weight groups within each row.

Within each cycle, four intermediate MAC output voltages V4, V3, V2, and V1 are generated, each representing the contribution from one 2-bit weight segment interacting with the corresponding 2-bit input segment. The capacitive network is structured such that:

- V1 (Group G0, bits [W1,W0]): No capacitive shift represents the LSB partial product.
- V2 (Group G1, bits [W3,W2]): Requires a 2-bit left shift (multiply by 4), implemented by a 4C capacitor.
- V3 (Group G2, bits [W5,W4]): Requires a 4-bit left shift (multiply by 16).
- V4 (Group G3, bits [W7,W6]): Requires a 6-bit left shift (multiply by 64).

Rather than implementing the full 6-bit capacitive shift (which would require a 64C capacitor), the shift-and-add process is partitioned into two hierarchical stages. In the first stage, V1 and V2 are summed to produce intermediate output Y0; concurrently, V3 and V4 are summed to produce Y1. In the second stage, Y1 is shifted by 4 bits (via a 16C scaling capacitor) and added to Y0, yielding the final single-cycle partial product P:

$$P = Y1 \times 2^4 + Y0$$

This two-stage hierarchical approach reduces the number of ADCs required from four to two per column one for digitizing Y0 and one for Y1 significantly reducing power, area, and design complexity.

4.9.4 Phase 4: Output Digitization

In the output phase, the two analog intermediate results Y_0 and Y_1 for each column are digitized by a pair of 8-bit Successive Approximation Register (SAR) ADCs. The SAR ADC is chosen for its energy efficiency and well-characterized performance in moderate-speed, moderate-resolution applications. The digitized outputs for cycle k are denoted Y_1 and Y_0 .

The digital partial sum P for each cycle is then computed digitally:

$$P = (Y_1 \ll 4) + Y_0$$

This digital combination recovers the full 8-bit precision of the partial MAC product for cycle.

4.10 Four-Cycle Accumulation and Final MAC Output

4.10.1 Partial Sums Across Cycles

Over the four computation cycles, four partial digital sums P_0 , P_1 , P_2 , and P_3 are obtained, corresponding to the four 2-bit input segments B_0 , B_1 , B_2 , and B_3 respectively. Each partial sum P_k captures the dot-product contribution of the k -th 2-bit input segment with the stored 8-bit weight vector.

The bit-significance of the four segments with respect to the 8-bit input requires that the partial sums be appropriately scaled before final accumulation:

- P_0 (from B_0 , bits [1:0]): No shift required LSB contribution.
- P_1 (from B_1 , bits [3:2]): Requires a 2-bit left shift (multiply by 4).
- P_2 (from B_2 , bits [5:4]): Requires a 4-bit left shift (multiply by 16).
- P_3 (from B_3 , bits [7:6]): Requires a 6-bit left shift (multiply by 64).

4.10.2 Final MAC Output

The final 8-bit MAC output for each column is obtained by digitally shifting and adding the four partial sums:

$$\text{MAC output} = P_0 + (P_1 \ll 2) + (P_2 \ll 4) + (P_3 \ll 6)$$

This digital accumulation is performed in a small digital post-processing block, completing the full 8-bit MAC computation. The result represents the inner product of the 8-bit input feature vector with the 8-bit weight vector stored in the corresponding column of the FeFET array.

4.11 ADC Count Comparison: Conventional vs. Proposed Approach

In a conventional multi-cycle analog CIM approach, each of the four 2-bit weight groups would produce an independent partial MAC output requiring its own ADC. For 4 partial outputs per

column across 8 columns, this would necessitate 32 ADCs in total for the 8×8 array a substantial hardware overhead.

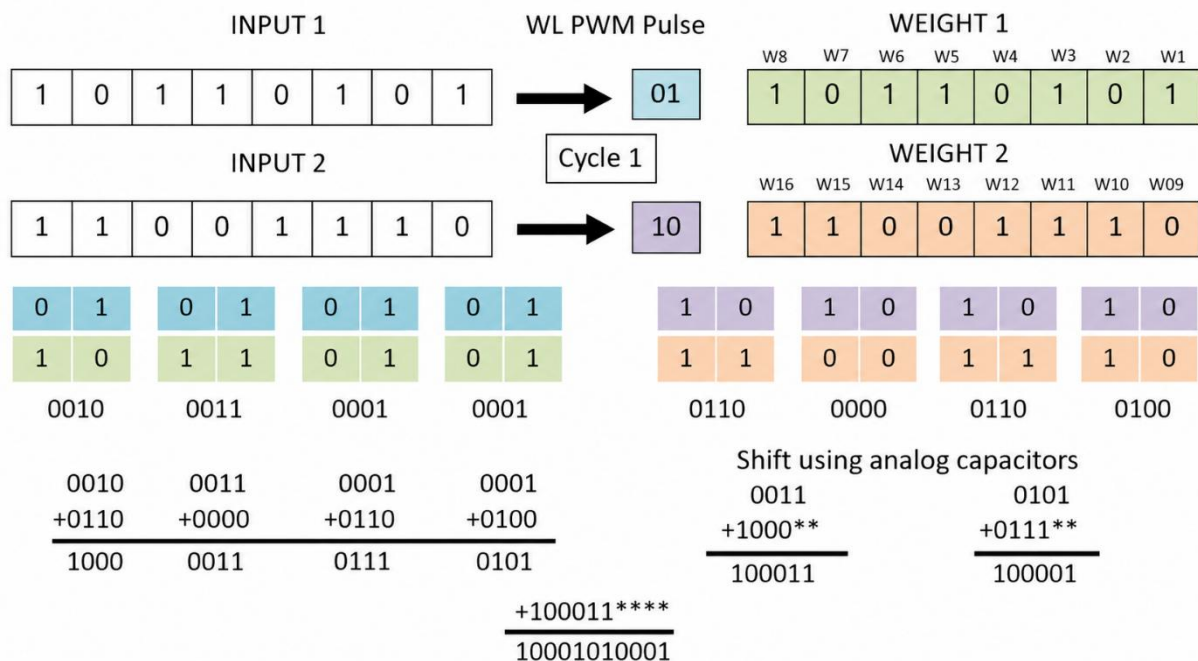
In the proposed FeFET architecture, the analog shift-and-add technique implemented by the hierarchical 1C-2C-4C capacitive network reduces the number of ADCs required from 4 to 2 per column. The two ADCs per column digitize the combined intermediate outputs Y0 and Y1, from which the full partial sum P_k is reconstructed digitally. For the 8-column array, only 16 ADCs are required in total, representing a 50% reduction compared to the conventional approach.

Architecture	ADCs/Column	Total ADCs (8 cols)	ADC Reduction
Conventional multi-cycle CIM	4	32	—
Proposed FeFET shift-and-add	2	16	50%

Table 4.3: ADC Count Comparison Conventional vs. Proposed Approach

This reduction in ADC count directly translates to lower power consumption and reduced silicon area, both critical for edge inference applications. The hierarchical capacitive shift-and-add thus provides a hardware-efficient path to 8-bit MAC accuracy in the analog domain.

4.12 Work Flow



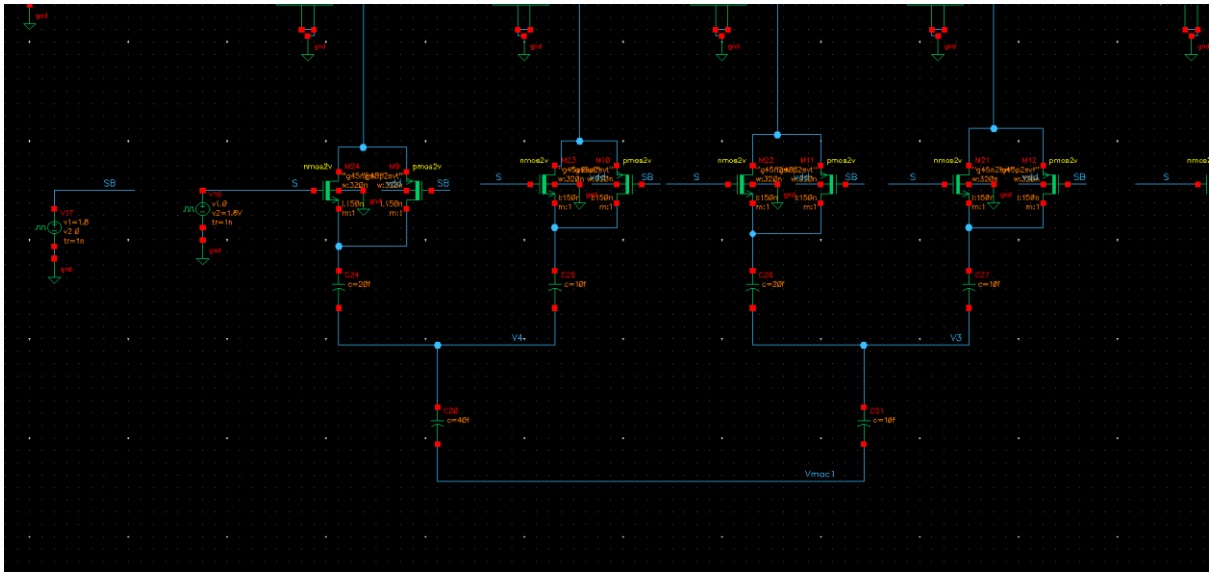


Figure4.6: Schematic of transmission gate 2c-1c and 4c-1c capacitive charge sharing network

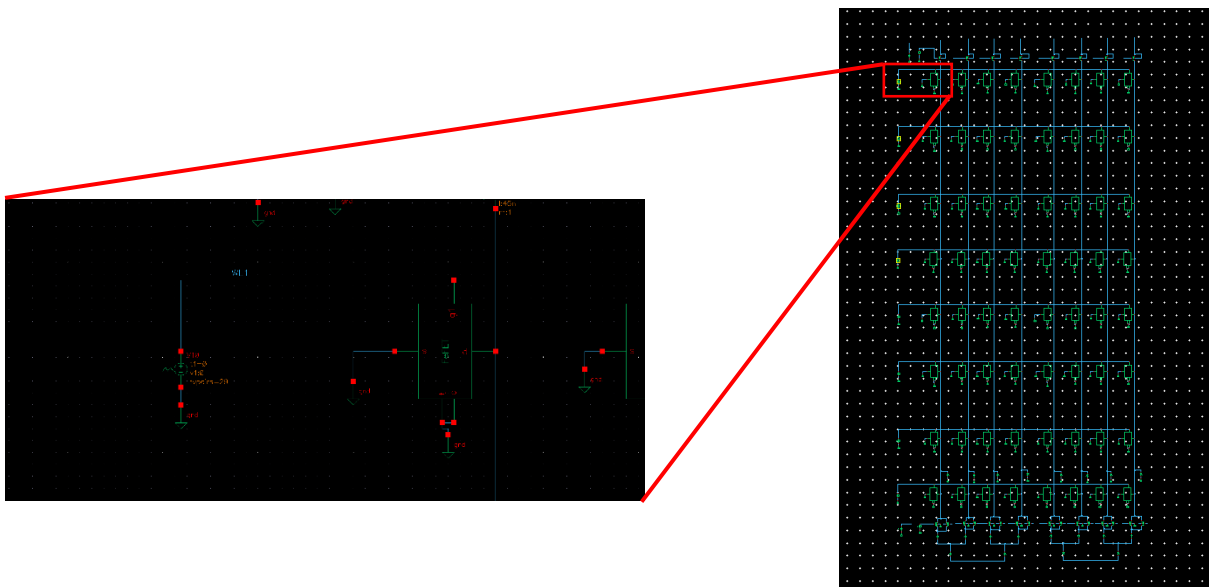


Figure4.7: Schematic of a 8×8 FeFET-Based Macro Array

4.13.2 Timing waveform

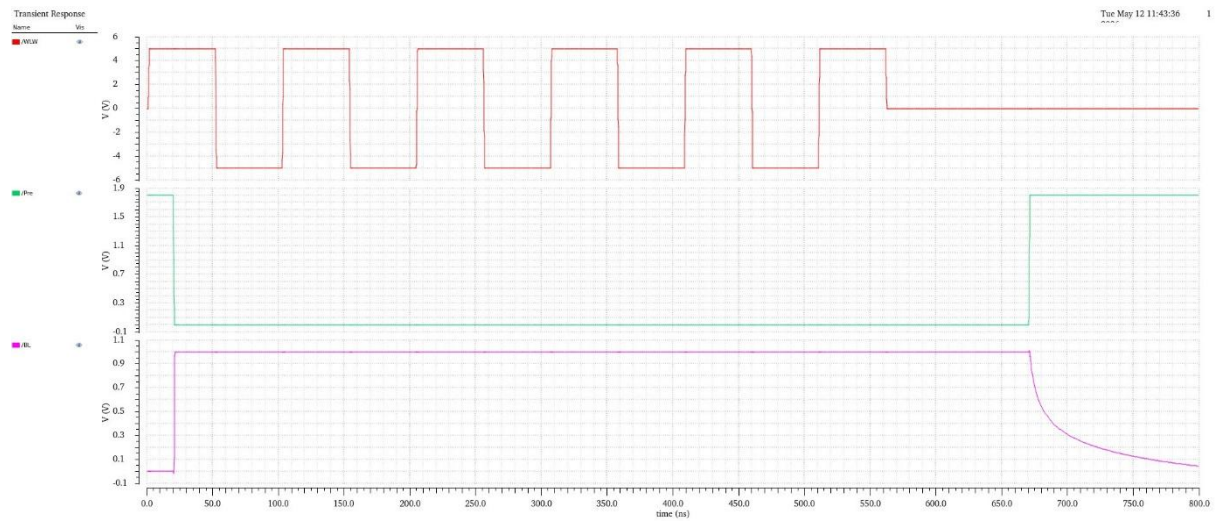


Figure4.8: Waveform of Write 1 operation in FeFET

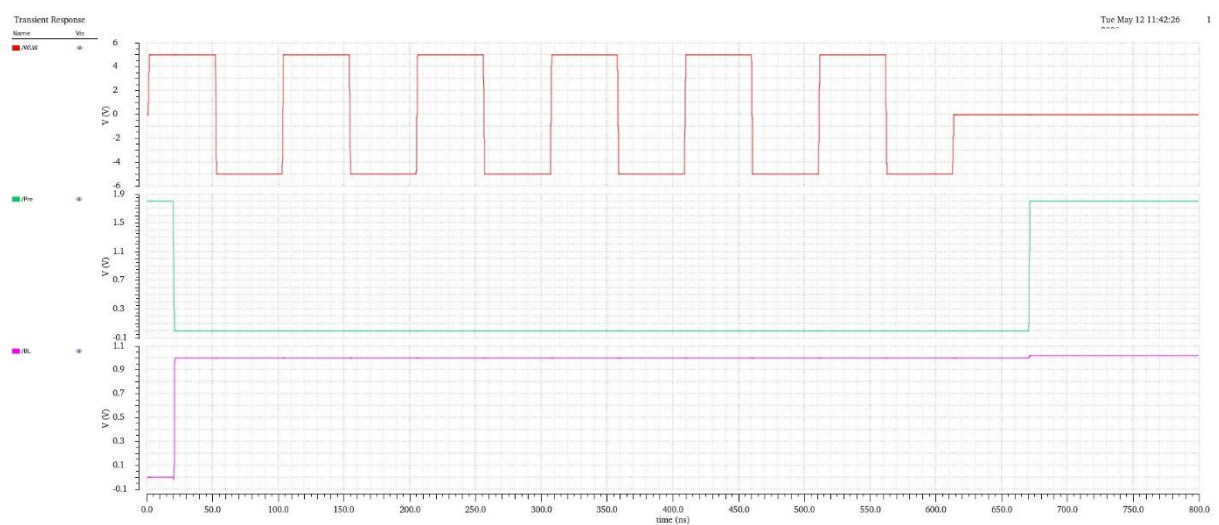


Figure4.9: Waveform of Write 0 operation in FeFET

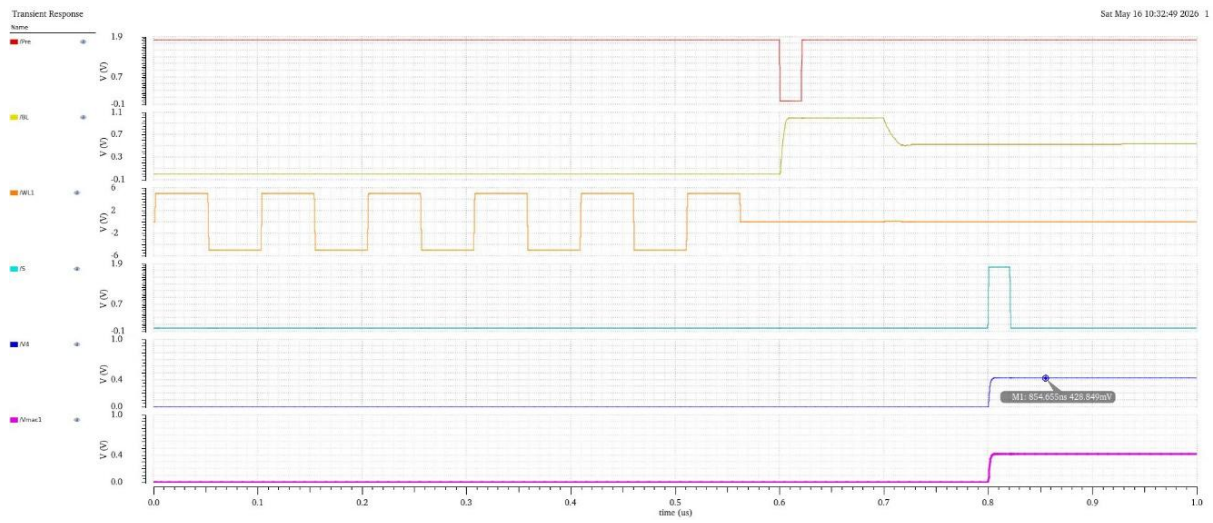


Figure4.10: Timing waveform for 8-bit MAC operation

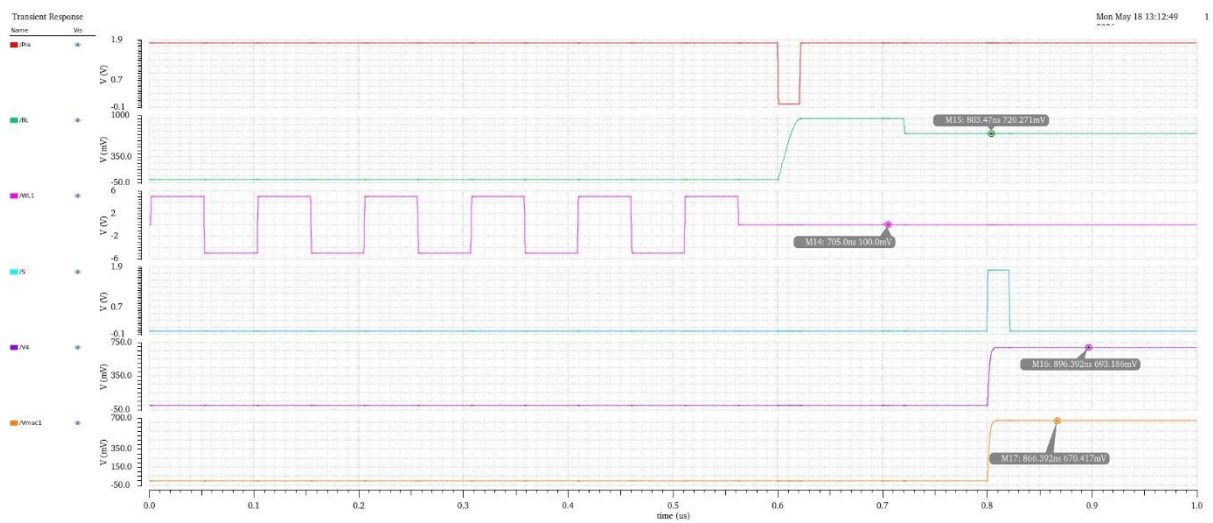


Figure4.11: Timing waveform for 8-bit MAC operation

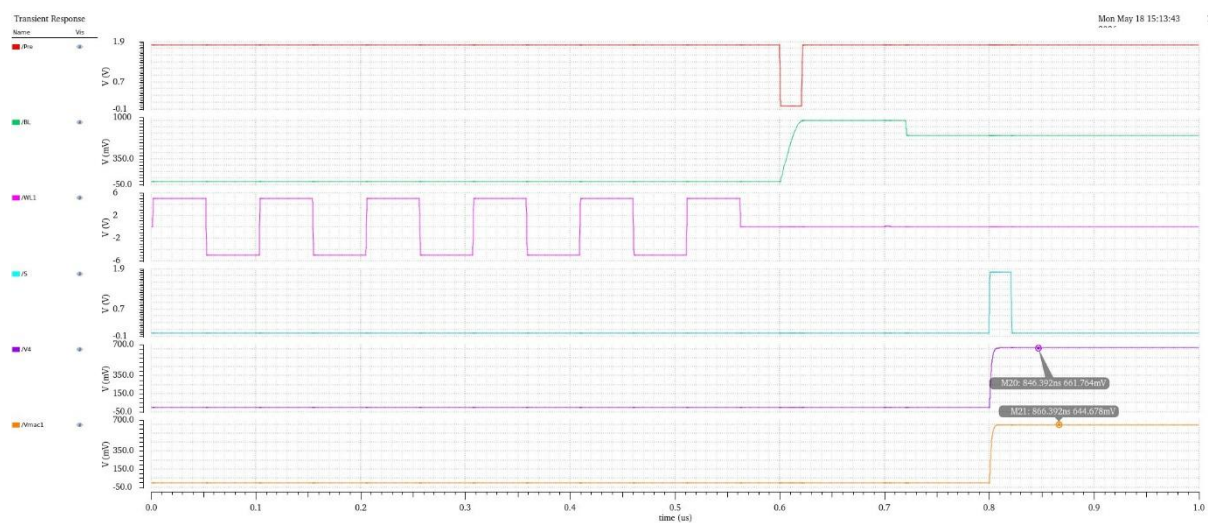


Figure4.12: Timing waveform for 8-bit MAC operation

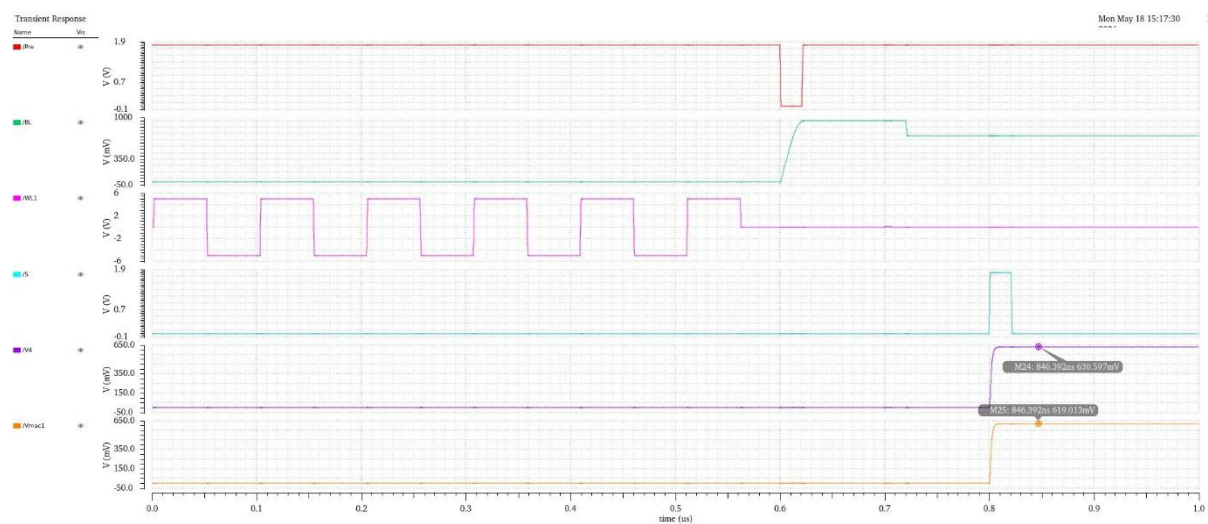


Figure4.13: Timing waveform for 8-bit MAC operation

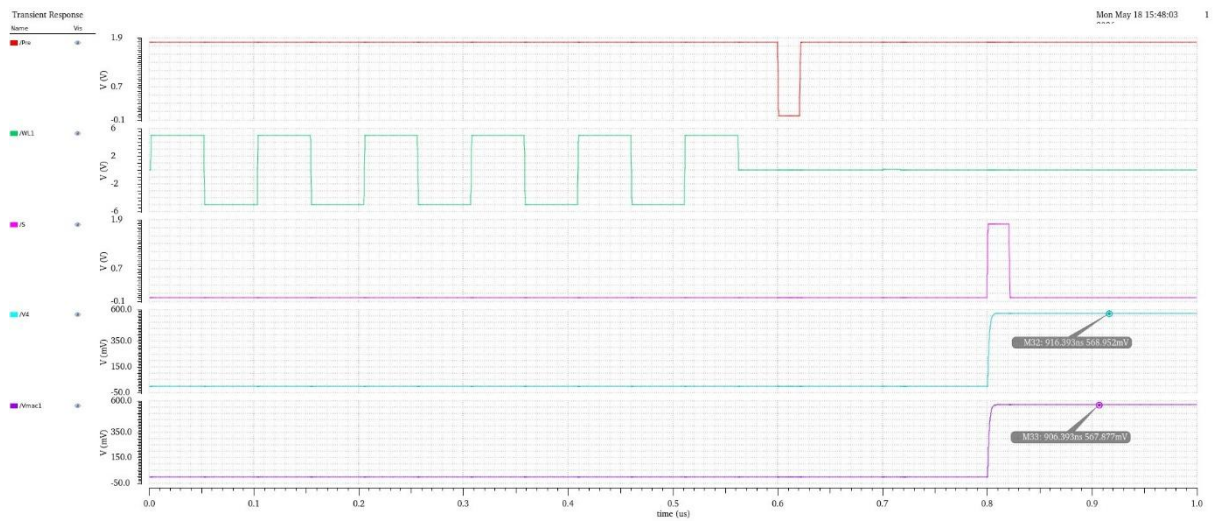


Figure4.14: Timing waveform for 8-bit MAC operation

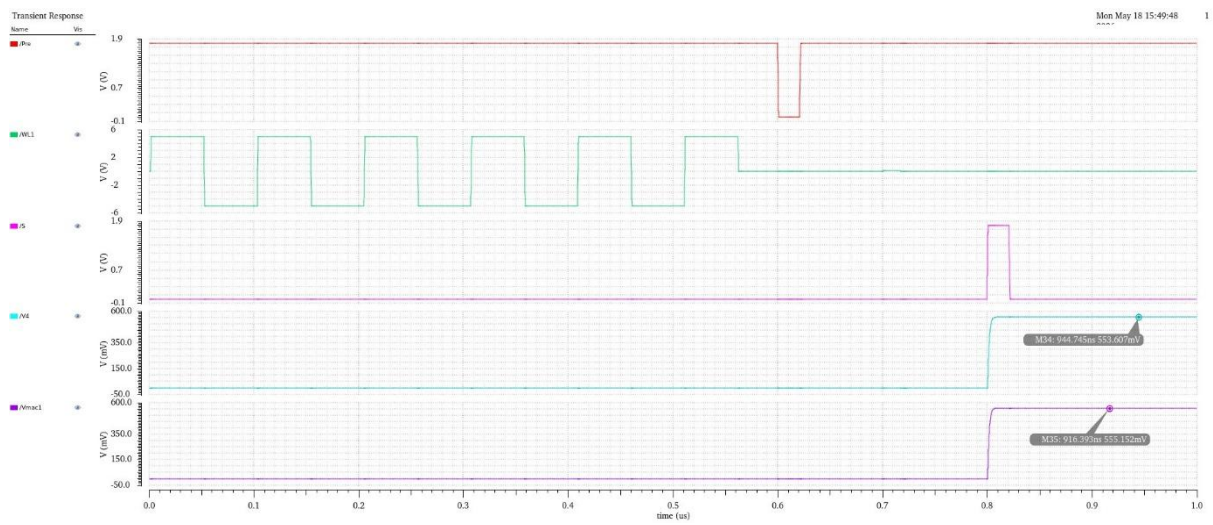


Figure4.15: Timing waveform for 8-bit MAC operation

4.13.3 Voltage Drop Graph

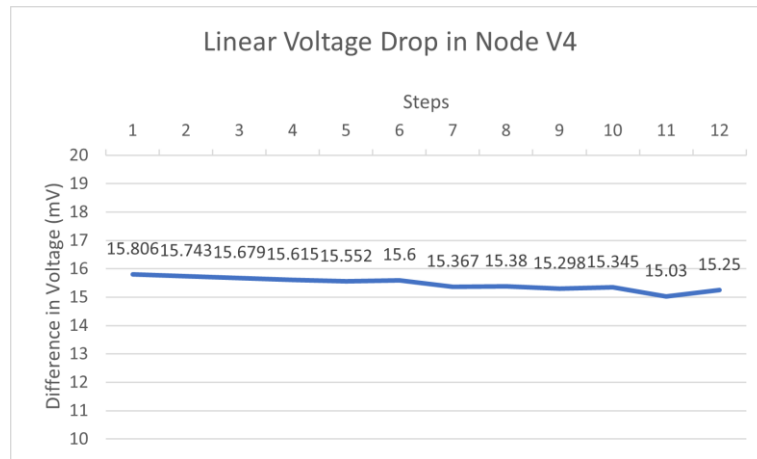


Figure4.16: Voltage Drop Graph of Node V4

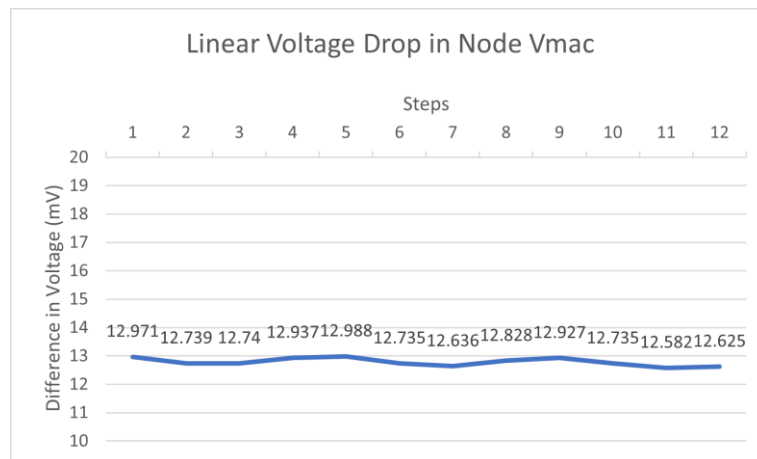


Figure4.17: Voltage Drop Graph of Node Vmac

4.13.4 PVT Analysis

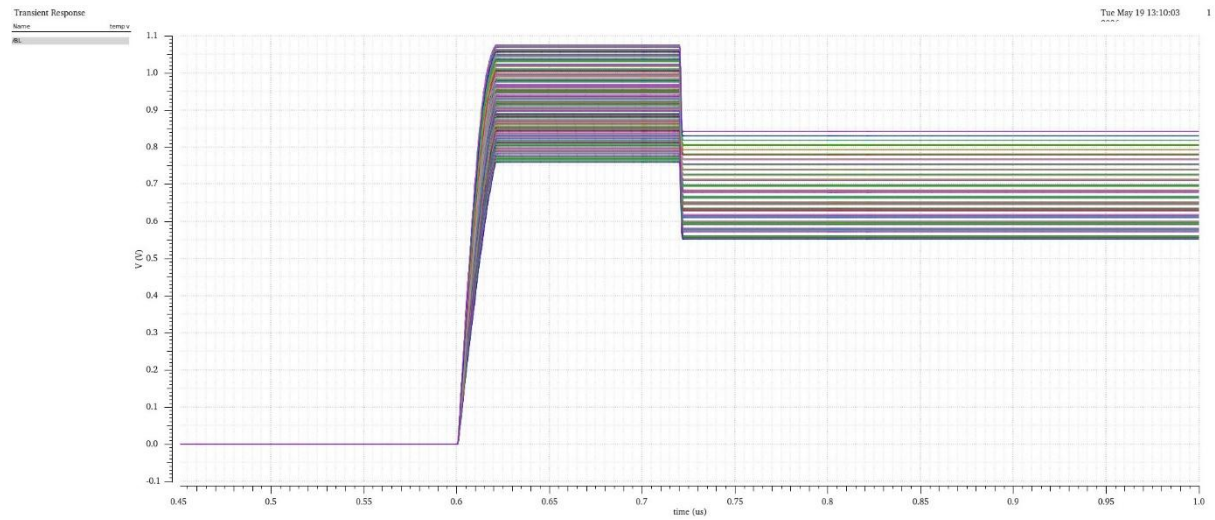


Figure 4.18: PVT Analysis BL

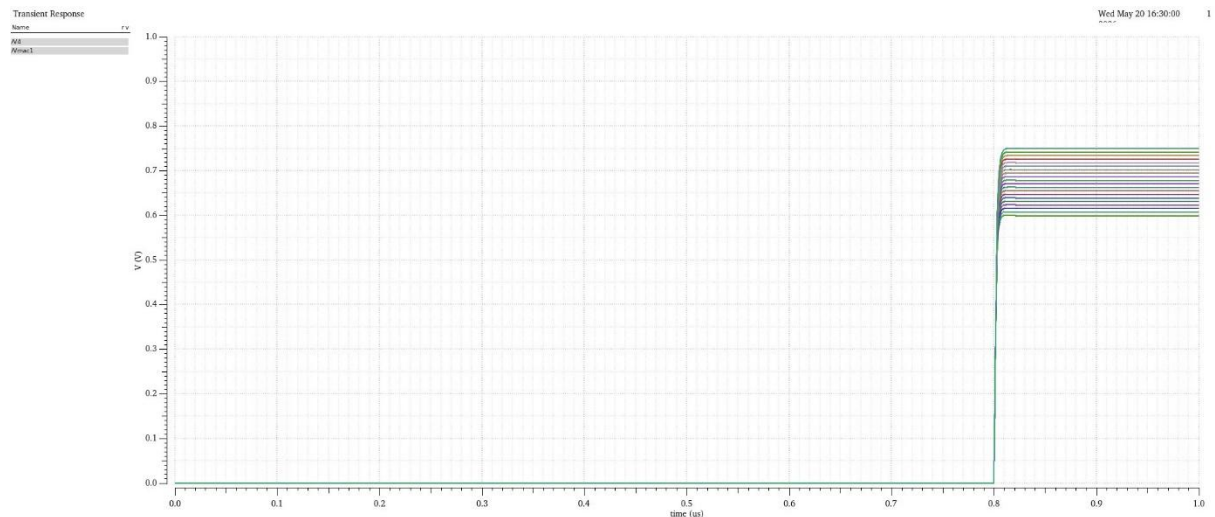


Figure 4.19: PVT Analysis V_4

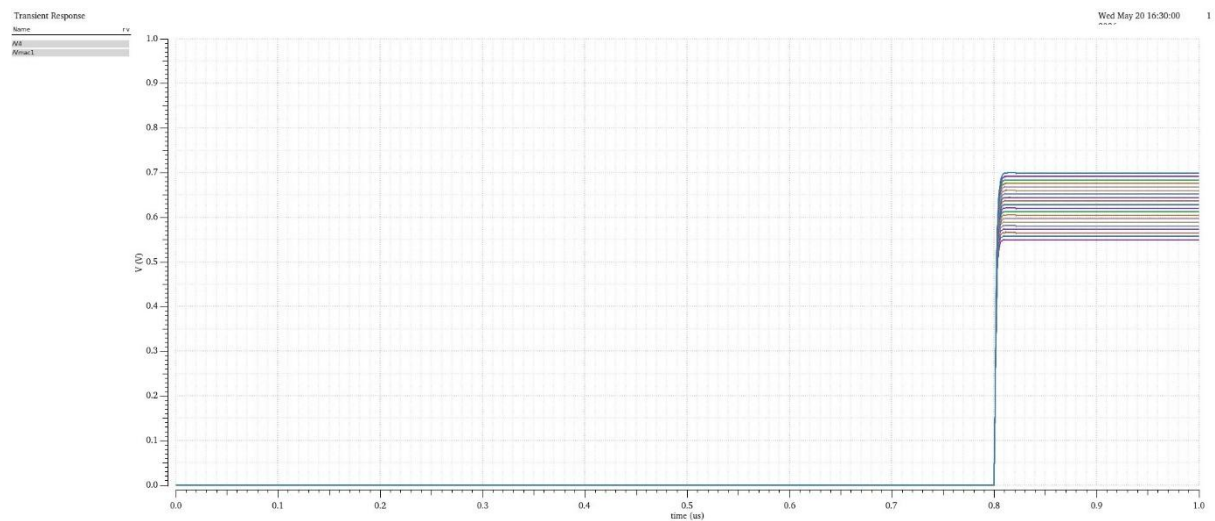


Figure 4.: PVT Analysis V_{mac}

4.14 Simulation Results

The proposed 8×8 FeFET CIM MAC architecture was simulated in Cadence Virtuoso under nominal conditions ($V_{DD} = 1.0$ V, 27°C , typical process corner). Key performance metrics are summarized below.

Performance Metric	This Work	Unit
Average Power Consumption	98	μW
Latency (per cycle)	16	ns
Throughput	2.13	GOPS
Energy Efficiency	8.59	TOPS/W
Supply Voltage	1.0	V
Array Size	8×8	FeFET cells
Bit Width	8	bits

Table 4.4: Key Performance Metrics of the Proposed FeFET CIM Array

The architecture achieves an energy efficiency of 8.59 TOPS/W at only $98 \mu\text{W}$ average power and 16 ns per-cycle latency, demonstrating significant improvement over conventional SRAM-based CIM designs. The 2.13 GOPS throughput is enabled by fully parallel row activation across all eight wordlines within each computation cycle.

References

- [1] J. von Neumann, "First Draft of a Report on the EDVAC," Moore School of Electrical Engineering, University of Pennsylvania, Tech. Rep., 1945.
- [2] W. A. Wulf and S. A. McKee, "Hitting the Memory Wall: Implications of the Obvious," ACM SIGARCH Computer Architecture News, vol. 23, no. 1, pp. 20–24, Mar. 1995.
- [3] A. Sebastian, M. Le Gallo, R. Khaddam-Aljameh, and E. Eleftheriou, "Memory Devices and Applications for In-Memory Computing," Nature Nanotechnology, vol. 15, pp. 529–544, 2020.
- [4] S. Yu, "Neuro-Inspired Computing With Emerging Nonvolatile Memorys," Proceedings of the IEEE, vol. 106, no. 2, pp. 260–285, Feb. 2018.
- [5] M. Imani, S. Gupta, Y. Kim, and T. Rosing, "FloatPIM: In-Memory Acceleration of Deep Neural Network Training with High Precision," in Proc. ACM/IEEE Design Automation Conference (DAC), 2019.
- [6] D. D. Lu, "Computing-in-memory with ferroelectric materials and beyond," Institute of Microelectronics, Dept. Elect. Eng., National Cheng Kung University, Tainan City, Taiwan, 2021.
- [7] H. Jiang, R. Liu, and S. Yu, "8T XNOR-SRAM based parallel compute-in-memory for deep neural network accelerator," in Proc. IEEE Int. Symp. Circuits Syst. (ISCAS), 2020, pp. 257–260.
- [8] A. Jaiswal, I. Chakraborty, A. Agrawal, and K. Roy, "8T SRAM Cell as a Multi-bit Dot Product Engine for Beyond von-Neumann Computing," IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 27, no. 11, pp. 2556–2567, Nov. 2019, doi: 10.1109/TVLSI.2019.2929245.
- [9] A. Biswas et al., "X-SRAM: Enabling in-Memory Boolean Computations in CMOS Static Random Access Memories," IEEE Trans. Circuits Syst. I, Reg. Papers, 2018.
- [10] H. Wang, R. Liu, R. Dorrance, D. Dasalukunte, D. Lake, and B. Carlton, "A Charge Domain SRAM Compute-in-Memory Macro with C-2C Ladder-Based 8-bit MAC Unit in 22-nm FinFET Process for Edge Inference," IEEE Journal of Solid-State Circuits, vol. 58, no. 4, pp. 1037–1050, Apr. 2023, doi: 10.1109/JSSC.2022.3232601.
- [11] Z. Lin, H. Zhan, Z. Chen, C. Peng, X. Wu, W. Lu, Q. Zhao, X. Li, and J. Chen, "Cascade Current Mirror to Improve Linearity and Consistency in SRAM In-Memory Computing," IEEE Journal of Solid-State Circuits, vol. 56, no. 8, pp. 2550–2564, Aug. 2021, doi: 10.1109/JSSC.2021.3063719.
- [12] M. Cao, N. Yu, H. Zhang, N. Lv, Y. Yang, and C. Han, "A High Linearity Current-Mirror MAC Circuit Design for SRAM Computing-in-Memory," in Proc. 9th Int. Conf. Integrated Circuits and Microsystems (ICICM), 2024, pp. 712–716, doi: 10.1109/ICICM63644.2024.10814160.
- [13] A. Goel, C. Singhal, D. Kushwaha, S. Mittal, and S. Dasgupta, "A 6T SRAM Analog CIM Macro for 8-bit MAC with Input/Weight Partitioning for High Signal Margin and Throughput," in Proc. IEEE Asia Pacific Conf. Circuits Syst. (APCCAS), 2025.

- [14] A. G. Amer, M. Ashok, X. Zhang, J. Cohn, and A. P. Chandrakasan, "A 14-nm Energy-Efficient and Reconfigurable Analog Current-Domain In-Memory Compute SRAM Accelerator," in Proc. IEEE Int. Conf. VLSI Design Embedded Syst. (VLSID), 2025, pp. 575–580.
- [15] C. Peng, J. Guo, S. Yan, Y. Wei, X. Chen, W. Lu, C. Dai, Z. Lin, and X. Wu, "A 28-nm Cascode Current Mirror-Based Inconsistency-Free Charging-and-Discharging SRAM-CIM Macro for High-Efficient Convolutional Neural Networks," IEEE Trans. Very Large Scale Integr. (VLSI) Syst., 2025, doi: 10.1109/TVLSI.2025.3552641.
- [16] K. Park, H. Jeong, S. Kim, J. Shin, M. Kim, and K. J. Lee, "A 701.7 TOPS/W Compute-in-Memory Processor with Time-Domain Computing for Spiking Neural Network," IEEE Trans. Circuits Syst. I, Reg. Papers, vol. 72, no. 1, pp. 25–38, Jan. 2025, doi: 10.1109/TCSI.2024.3480350.
- [17] I. Kataeva, F. Merrih-Bayat, E. Merrih-Bayat, and B. Linares-Barranco, "Efficient Training Algorithms for Neural Networks Based on Memristive Crossbar Networks," in Proc. IEEE International Joint Conference on Neural Networks (IJCNN), 2015.
- [18] B. Razavi, Design of Analog CMOS Integrated Circuits, 2nd ed. New York, NY, USA: McGraw-Hill, 2017.
- [19] S. S. Cheema et al., "Ultrathin Ferroic HfO₂-ZrO₂ Superlattice Gate Stack for Advanced Transistors," Nature, vol. 604, pp. 65–71, 2022.
- [20] T. Kämpfe, P. Lehnen, S. Abdulazhanov, P. Bruns, M. Trentzsch, S. Dünkel, S. Beyer, T. Mikolajick, and U. Schroeder, "Ferroelectric HfO₂ Films in Analog Neuromorphic Computing Applications," in Proc. Non-Volatile Memory Technology Symposium (NVMTS), 2019.
- [21] M. Jerry, P.-J. Chen, J. Zhang, P. Sharma, K. Ni, S. Yu, and S. Datta, "Ferroelectric FET Analog Synapse for Acceleration of Deep Neural Network Training," in Proc. IEEE International Electron Devices Meeting (IEDM), 2017, pp. 6.2.1–6.2.4.
- [22] K. Ni, J. Smith, B. Gao, L. Yin, X. Li, M. Jerry, J. A. Smith, M. Takamura, F. Rao, H. Wu, H. Qian, and S. Datta, "In-Memory Computing Primitive for Sensor Data Fusion in 28 nm HKMG FeFET Technology," in Proc. IEEE International Electron Devices Meeting (IEDM), 2018, pp. 16.3.1–16.3.4.
- [23] X. Yin, A. Aziz, J. Nahas, S. Datta, S. Gupta, M. Niemier, and X. S. Hu, "Exploiting Ferroelectric FETs for Low-Power Non-Volatile Logic-in-Memory Circuits," 2018.
- [24] M. Mulaosmanovic et al., "First Demonstration of Ultra-High Precision 4Kb 28nm HKMG FeFET-1T Based Memory Array Macro for Highly Scaled Deep Learning Applications," in Proc. IEEE International Electron Devices Meeting (IEDM), 2021, pp. 26.5.1–26.5.4, doi: 10.1109/IEDM19574.2021.9720633.
- [25] S. De, S. Dunkel, J. Pelloux-Prayer, L. Grenouillet, J. Borrel, N. Bernier, G. Audoit, L. Baines, P. Chiquet, F. Gaillard, B. De Salvo, and E. Nowak, "Demonstration of Linear Multiply-Accumulate Operation with 28 nm High-K Metal Gate FeFET Crossbar Array," IEEE Electron Device Letters, vol. 43, no. 12, pp. 2030–2033, Dec. 2022.
- [26] S. De, S. Dunkel, J. Pelloux-Prayer, L. Grenouillet, J. Borrel, N. Bernier, G. Audoit, F. Gaillard, B. De Salvo, and E. Nowak, "28 nm HKMG-Based Current Limited FeFET

- Crossbar Array for Inference Application," IEEE Transactions on Electron Devices, vol. 69, no. 12, pp. 6719–6726, Dec. 2022.
- [27] X. Lu, S. Chen, Z. Pang, S. Sun, X. Yin, and C. Zhuo, "In-Memory Multi-Bit Multiplication and Accumulation (MAC) Using FeFET for Energy Efficient IoT," in Proc. 2nd Int. Conf. Intelligent Technology and Embedded Systems (ICITES), 2022, pp. 27–33, doi: 10.1109/ICITES56274.2022.9943617.
 - [28] N. Laleni and T. Kämpfe, "An Efficient FeFET-Based CIM Macro Readout Inspired by the 3T Pixel Array Cell," in Proc. IEEE International Conference on Synthesis, Modeling, Analysis and Simulation Methods and Applications to Circuit Design (SMACD), 2024.
 - [29] S. Garg, Z. Jia, A. Phadke, and H. Yu, "A 28 nm FeFET Compute-in-Memory Macro with 64×64 Array Size and On-Chip 4-Bit Flash ADC," IEEE Solid-State Circuits Letters, vol. 9, 2026.
 - [30] X. Yin et al., "A 28-nm FeFET Compute-in-Memory Macro With 64×64 Array Size and On-Chip 4-Bit Flash ADC," IEEE Journal of Solid-State Circuits, vol. 58, no. 4, pp. 1084–1093, Apr. 2023, doi: 10.1109/JSSC.2022.3232477.